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1 PDF matching conditions

If the (Zero Mass) Variable Flavour Number Scheme (ZM-VFNS) at NNLO is considered, matching conditions for the PDFs and the coupling constant at the heavy-quarks thresholds (m_c^2 , m_b^2 , and m_t^2) must be implemented. This is due to the fact that we are working with an “effective theory” where before a certain threshold, say m_h^2 , the heavy-quark flavour h is treated as infinitely massive, while after the crossing of the same threshold the same flavour is treated as massless. This results in discontinuities of PDFs and coupling constant in correspondence of the thresholds. Such discontinuities can be evaluated in perturbation theory [1].

The discontinuity of the PDF l (in Mellin space) of a light quark⁽¹⁾ just beyond the threshold m_h^2 ($= m_c^2, m_b^2, m_t^2$), where the effective flavour number goes from n_f to $n_f + 1$, is given as a function of the same PDF just before the threshold by the following relation [2]:

$$l^{(n_f+1)}(N, m_h^2) = [1 + a_s^2(m_h^2)A_{qq,h}^{NS,(2)}(N)]l^{(n_f)}(N, m_h^2). \quad (1)$$

with $l = u, \bar{u}, d, \bar{d}, \dots$, while the gluon distribution function is given by:

$$g^{(n_f+1)}(N, m_h^2) = [1 + a_s^2(m_h^2)A_{gg,h}^{S,(2)}(N)]g^{(n_f)}(N, m_h^2) + a_s^2(m_h^2)A_{gg,h}^{S,(2)}(N)\Sigma^{(n_f)}(N, m_h^2) \quad (2)$$

and, in the end, the sum of heavy quark h and its anti-quark $\bar{h}$, which are going to be produced after the threshold m_h^2 , is:

$$(h^{(n_f+1)} + \bar{h}^{(n_f+1)})(N, m_h^2) = a_s^2(m_h^2)[\tilde{A}_{hq}^{S,(2)}(N)\Sigma^{(n_f)}(N, m_h^2) + \tilde{A}_{hg}^{S,(2)}(N)g^{(n_f)}(N, m_h^2)]. \quad (3)$$

Of course, we have $h = \bar{h}$.

Now, since:

$$\Sigma^{(n_f+1)} = \sum_{l=1}^{n_f} (l^{(n_f+1)} + \bar{l}^{(n_f+1)}) + (h^{(n_f+1)} + \bar{h}^{(n_f+1)}) \quad (4)$$

we find that the matching condition for the singlet is:

$$\begin{aligned} \Sigma^{(n_f+1)}(N, m_h^2) &= [1 + a_s^2(m_h^2)A_{qq,h}^{NS,(2)}(N)]\Sigma^{(n_f)}(N, m_h^2) + \\ &a_s^2(m_h^2)[\tilde{A}_{hq}^{S,(2)}(N)\Sigma^{(n_f)}(N, m_h^2) + \tilde{A}_{hg}^{S,(2)}(N)g^{(n_f)}(N, m_h^2)] \end{aligned} \quad (5)$$

so that, from eqs. (2) and (5):

$$\begin{aligned} \begin{pmatrix} \Sigma^{(n_f+1)} \\ g^{(n_f+1)} \end{pmatrix} &= \begin{pmatrix} 1 + a_s^2[A_{qq,h}^{NS,(2)} + \tilde{A}_{hq}^{S,(2)}] & a_s^2\tilde{A}_{hg}^{S,(2)} \\ a_s^2\tilde{A}_{hq}^{S,(2)} & 1 + a_s^2\tilde{A}_{gg,h}^{S,(2)} \end{pmatrix} \begin{pmatrix} \Sigma^{(n_f)} \\ g^{(n_f)} \end{pmatrix} \\ &= \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_s^2 A_{qq,h}^{NS,(2)} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + a_s^2 \begin{pmatrix} \tilde{A}_{hq}^{S,(2)} & \tilde{A}_{hg}^{S,(2)} \\ \tilde{A}_{gg,h}^{S,(2)} & \tilde{A}_{gg,h}^{S,(2)} \end{pmatrix} \right] \begin{pmatrix} \Sigma^{(n_f)} \\ g^{(n_f)} \end{pmatrix} \end{aligned} \quad (6)$$

where we have omitted all the dependencies. So we have obtained the matching conditions for the singlet and the gluon distribution functions.

Now we consider the other distributions. The valence distribution V for $n_f + 1$ active (light) flavours just beyond the threshold m_h^2 is defined as:

$$V^{(n_f+1)} = \sum_{l=1}^{n_f} (l^{(n_f+1)} - \bar{l}^{(n_f+1)}) + (h^{(n_f+1)} - \bar{h}^{(n_f+1)}) \quad (7)$$

¹Note that the light quarks run from 1 to n_f .

but, since $h = \bar{h}$, the last term vanish and we are left with:

$$V^{(n_f+1)} = \sum_{l=1}^{n_f} (l^{(n_f+1)} - \bar{l}^{(n_f+1)}) =$$

$$[1 + a_s^2 A_{qq,h}^{NS,(2)}] \sum_{l=1}^{n_f} (l^{(n_f)} - \bar{l}^{(n_f)}) = [1 + a_s^2 A_{qq,h}^{NS,(2)}] V^{(n_f)} . \quad (8)$$

So, this is the matching condition for the valence distribution V .

Now we consider the valence distribution V_3 and V_8 which are both composed only by light quarks, namely:

$$V_3 = (u - \bar{u}) - (d - \bar{d}) \quad \text{and} \quad V_8 = (u - \bar{u}) + (d - \bar{d}) - 2(s - \bar{s}) \quad (9)$$

so that, in these cases, the matching conditions work as in the case of V , i.e.:

$$V_{3,8}^{(n_f+1)} = [1 + a_s^2 A_{qq,h}^{NS,(2)}] V_{3,8}^{(n_f)} . \quad (10)$$

The same holds for T_3 and T_8 , which are defined as:

$$T_3 = (u + \bar{u}) - (d + \bar{d}) \quad \text{and} \quad V_8 = (u + \bar{u}) + (d + \bar{d}) - 2(s + \bar{s}) \quad (11)$$

so:

$$T_{3,8}^{(n_f+1)} = [1 + a_s^2 A_{qq,h}^{NS,(2)}] T_{3,8}^{(n_f)} . \quad (12)$$

The remaining valence distribution V_{15} , V_{24} and V_{35} are defined as:

$$\begin{aligned} V_{15} &= (u - \bar{u}) + (d - \bar{d}) + (s - \bar{s}) - 3(c - \bar{c}) \\ V_{24} &= (u - \bar{u}) + (d - \bar{d}) + (s - \bar{s}) + (c - \bar{c}) - 4(b - \bar{b}) \\ V_{35} &= (u - \bar{u}) + (d - \bar{d}) + (s - \bar{s}) + (c - \bar{c}) + (b - \bar{b}) - 5(t - \bar{t}) \end{aligned} \quad (13)$$

and since in each one of them the heavy quarks appear always as difference between quark and anti-quark, they cancel exactly. For example, at the m_b^2 threshold, V_{15} is entirely composed by light quarks so there is no problem, while V_{24} and V_{35} have also a b -quark contribution, given by $-4(b - \bar{b})$ and $(b - \bar{b})$ respectively (of course, the $t(\bar{t})$ distribution is zero). Anyway, this terms give no matching condition since the b contribution is exactly equal to the $\bar{b}$ contribution, so that they cancel. So:

$$V_{15,24,35}^{(n_f+1)} = [1 + a_s^2 A_{qq,h}^{NS,(2)}] V_{15,24,35}^{(n_f)} . \quad (14)$$

In the end, to deal with T_{15} , T_{24} and T_{35} , we have to specify the threshold. Indeed, in these cases the heavy quark contribution does not cancel. Their definition is:

$$\begin{aligned} T_{15} &= (u + \bar{u}) + (d + \bar{d}) + (s + \bar{s}) - 3(c + \bar{c}) \\ T_{24} &= (u + \bar{u}) + (d + \bar{d}) + (s + \bar{s}) + (c + \bar{c}) - 4(b + \bar{b}) \\ T_{35} &= (u + \bar{u}) + (d + \bar{d}) + (s + \bar{s}) + (c + \bar{c}) + (b + \bar{b}) - 5(t + \bar{t}) . \end{aligned} \quad (15)$$

Just before the threshold m_c^2 we have only 3 active light flavours (u , d and s), while just beyond m_c^2 we have 4 active flavours and among them the flavour c is considered to be heavy. Of course, we have no b and t contribution (so T_{24} and T_{35} are equal). So the matching conditions are:

$$T_{15}^{(4)} = [1 + a_s^2 A_{qq,c}^{NS,(2)}] \underbrace{\sum_{l=u,d,s} (l^{(3)} + \bar{l}^{(3)})}_{\Sigma^{(3)}} - 3a_s^2 [\tilde{A}_{cq}^{S,(2)} \Sigma^{(3)} + \tilde{A}_{cg}^{S,(2)} g^{(3)}] =$$

$$\left(1 + a_s^2 [A_{qq,c}^{NS,(2)} - 3\tilde{A}_{cq}^{S,(2)}] \quad -3a_s^2 \tilde{A}_{cg}^{S,(2)} \right) \begin{pmatrix} \Sigma^{(3)} \\ g^{(3)} \end{pmatrix} \quad (16)$$

while:

$$T_{24,35}^{(4)} = \begin{pmatrix} 1 + a_s^2[A_{qq,c}^{NS,(2)} + \tilde{A}_{cq}^{S,(2)}] & a_s^2\tilde{A}_{cg}^{S,(2)} \end{pmatrix} \begin{pmatrix} \Sigma^{(3)} \\ g^{(3)} \end{pmatrix}. \quad (17)$$

We can put the above relation in a matricial form:

$$\begin{pmatrix} T_{15}^{(4)} \\ T_{24}^{(4)} \\ T_{35}^{(4)} \end{pmatrix} = \begin{pmatrix} 1 + a_s^2[A_{qq,c}^{NS,(2)} - 3\tilde{A}_{cq}^{S,(2)}] & -3a_s^2\tilde{A}_{cg}^{S,(2)} \\ 1 + a_s^2[A_{qq,c}^{NS,(2)} + \tilde{A}_{cq}^{S,(2)}] & a_s^2\tilde{A}_{cg}^{S,(2)} \\ 1 + a_s^2[A_{qq,c}^{NS,(2)} + \tilde{A}_{cq}^{S,(2)}] & a_s^2\tilde{A}_{cg}^{S,(2)} \end{pmatrix} \begin{pmatrix} \Sigma^{(3)} \\ g^{(3)} \end{pmatrix}. \quad (18)$$

But now it is easy to generalize. At m_b^2 , T_{15} does not contain, so:

$$T_{15}^{(5)} = [1 + a_s^2 A_{qq,b}^{NS,(2)}] T_{15}^{(4)} \quad (19)$$

while:

$$\begin{pmatrix} T_{24}^{(5)} \\ T_{35}^{(5)} \end{pmatrix} = \begin{pmatrix} 1 + a_s^2[A_{qq,b}^{NS,(2)} - 4\tilde{A}_{bq}^{S,(2)}] & -4a_s^2\tilde{A}_{bg}^{S,(2)} \\ 1 + a_s^2[A_{qq,b}^{NS,(2)} + \tilde{A}_{bq}^{S,(2)}] & a_s^2\tilde{A}_{bg}^{S,(2)} \end{pmatrix} \begin{pmatrix} \Sigma^{(4)} \\ g^{(4)} \end{pmatrix}. \quad (20)$$

Finally, at m_t^2 we have:

$$T_{15}^{(6)} = [1 + a_s^2 A_{qq,t}^{NS,(2)}] T_{15}^{(5)} \quad (21)$$

$$T_{24}^{(6)} = [1 + a_s^2 A_{qq,t}^{NS,(2)}] T_{25}^{(5)}$$

and:

$$T_{35}^{(6)} = \begin{pmatrix} 1 + a_s^2[A_{qq,t}^{NS,(2)} - 5\tilde{A}_{tq}^{S,(2)}] & -5a_s^2\tilde{A}_{tg}^{S,(2)} \end{pmatrix} \begin{pmatrix} \Sigma^{(5)} \\ g^{(5)} \end{pmatrix}. \quad (22)$$

An explicit calculation for the coefficients $A^{(2)}$ in the x -space can be found in [hep-ph/9612398]. Anyhow, that calculation is performed more generally in the case $m_h^2 \neq \mu_F^2$. This results in extra-terms proportional to $\ln(m_h^2/\mu_F^2)$, which vanish if, as we do, one takes the factorisation scale μ^2 coinciding with the scale of the process Q^2 . Moreover, in that case also NLO ($\propto a_s$) appear in the matching conditions.

To summarise the PDF matching conditions at the threshold m_h^2 , we have that:

- singlet and gluon couple as follows:

$$\begin{pmatrix} \Sigma^{(n_f+1)} \\ g^{(n_f+1)} \end{pmatrix} = \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_s^2 A_{qq,h}^{NS,(2)} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + a_s^2 \begin{pmatrix} \tilde{A}_{hq}^{S,(2)} & \tilde{A}_{hg}^{S,(2)} \\ A_{gg,h}^{S,(2)} & A_{gg,h}^{S,(2)} \end{pmatrix} \right] \begin{pmatrix} \Sigma^{(n_f)} \\ g^{(n_f)} \end{pmatrix}, \quad (23)$$

- from eqs. (8), (10), (14) and (12), one can see that V , $V_{3,8,\dots,35}$ and $T_{3,8}$ behave in the same way, i.e.:

$$P^{(n_f+1)} = [1 + a_s^2 A_{qq,h}^{NS,(2)}] P^{(n_f)} \quad \text{with} \quad P = V, V_3, \dots, V_{35}, T_3, T_8, \quad (24)$$

- T_{15} , T_{24} and T_{35} have different matching conditions depending on the threshold. In particular: for $m_h^2 = m_c^2$ they are given by eq. (18), for $m_h^2 = m_b^2$ they are given by eqs. (19) and (20) and for $m_h^2 = m_t^2$ they are given by eqs. (21) and (22)

In the following Sections we will discuss how to write the evolution kernels in the presence of the matching conditions. We will explicitly consider only the forward evolution, i.e. the final scale Q^2 greater than the initial one Q_0^2 . Anyway the backward evolution ($Q_0^2 > Q^2$) can be easily obtained from the forward one. In fact, given the evolution kernel Γ , the following relation holds:

$$\Gamma(Q^2, Q_0^2) \Gamma(Q_0^2, Q^2) = 1 \implies \Gamma(Q^2, Q_0^2) = \Gamma^{-1}(Q_0^2, Q^2). \quad (25)$$

so, if $Q^2 > Q_0^2$ the code computes directly $\Gamma(Q^2, Q_0^2)$, else if $Q_0^2 > Q^2$ the code evaluates first the forward evolution $\Gamma(Q_0^2, Q^2)$ and then, to get $\Gamma(Q^2, Q_0^2)$, it calculates $\Gamma^{-1}(Q_0^2, Q^2)$.

1.1 Matching Conditions on the Evolution Kernels: 0 Thresholds Crossing

Before to discuss the crossing of the thresholds, it would be useful to write down the evolution kernels in the “trivial” situation of no threshold crossing. There are 4 particular cases: 1) $Q_0^2 < Q^2 < m_c^2$ with $n_f = 3$ active flavours, 2) $m_c^2 < Q_0^2 < Q^2 < m_b^2$ with $n_f = 4$ active flavours, 3) $m_b^2 < Q_0^2 < Q^2 < m_t^2$ with $n_f = 5$ active flavours and 4) $m_t^2 < Q_0^2 < Q^2$ with $n_f = 6$ active flavours, which do not need the introduction of the matching conditions. So, in the following Subsections we will write down the evolution of the whole PDF set the these cases.

1.1.1 $Q_0^2 < Q^2 < m_c^2$

- **Singlet and gluon:**

$$\begin{pmatrix} \Sigma^{(3)}(Q^2) \\ g^{(3)}(Q^2) \end{pmatrix} = \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, Q_0^2)} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (26)$$

- **V:**

$$V^{(3)}(Q^2) = \Gamma^v(Q^2, Q_0^2) V^{(3)}(Q_0^2) \quad (27)$$

- **V₃ and V₈:**

$$V_{3,8}^{(3)}(Q^2) = \Gamma^-(Q^2, Q_0^2) V_{3,8}^{(3)}(Q_0^2) \quad (28)$$

- **V₁₅, V₂₄ and V₃₅:**

$$V_{15,24,35}^{(3)}(Q^2) = \Gamma^v(Q^2, Q_0^2) V^{(3)}(Q_0^2) \quad (29)$$

- **T₃ and T₈:**

$$T_{3,8}^{(3)}(Q^2) = \Gamma^+(Q^2, Q_0^2) T_{3,8}^{(3)}(Q_0^2) \quad (30)$$

- **T₁₅, T₂₄ and T₃₅:**

$$T_{15,24,35}^{(3)}(Q^2) = \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, Q_0^2)} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (31)$$

1.1.2 $m_c^2 < Q_0^2 < Q^2 < m_b^2$

- **Singlet and gluon:**

$$\begin{pmatrix} \Sigma^{(4)}(Q^2) \\ g^{(4)}(Q^2) \end{pmatrix} = \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, Q_0^2)} \begin{pmatrix} \Sigma^{(4)}(Q_0^2) \\ g^{(4)}(Q_0^2) \end{pmatrix} \quad (32)$$

- **V:**

$$V^{(4)}(Q^2) = \Gamma^v(Q^2, Q_0^2) V^{(4)}(Q_0^2) \quad (33)$$

- **V₃, V₈ and V₁₅:**

$$V_{3,8,15}^{(4)}(Q^2) = \Gamma^-(Q^2, Q_0^2) V_{3,8,15}^{(4)}(Q_0^2) \quad (34)$$

- **V₂₄ and V₃₅:**

$$V_{24,35}^{(4)}(Q^2) = \Gamma^v(Q^2, Q_0^2) V^{(4)}(Q_0^2) \quad (35)$$

- **T₃, T₈ and T₁₅:**

$$T_{3,8,15}^{(4)}(Q^2) = \Gamma^+(Q^2, Q_0^2) T_{3,8,15}^{(4)}(Q_0^2) \quad (36)$$

- **T₂₄ and T₃₅:**

$$T_{24,35}^{(4)}(Q^2) = \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, Q_0^2)} \begin{pmatrix} \Sigma^{(4)}(Q_0^2) \\ g^{(4)}(Q_0^2) \end{pmatrix} \quad (37)$$

1.1.3 $m_b^2 < Q_0^2 < Q^2 < m_t^2$

- **Singlet and gluon:**

$$\begin{pmatrix} \Sigma^{(5)}(Q^2) \\ g^{(5)}(Q^2) \end{pmatrix} = \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, Q_0^2)} \begin{pmatrix} \Sigma^{(5)}(Q_0^2) \\ g^{(5)}(Q_0^2) \end{pmatrix} \quad (38)$$

- **V:**

$$V^{(5)}(Q^2) = \Gamma^v(Q^2, Q_0^2) V^{(5)}(Q_0^2) \quad (39)$$

- **V₃, V₈, V₁₅ and V₂₄:**

$$V_{3,8,15,24}^{(5)}(Q^2) = \Gamma^-(Q^2, Q_0^2) V_{3,8,15,24}^{(5)}(Q_0^2) \quad (40)$$

- **V₃₅:**

$$V_{35}^{(5)}(Q^2) = \Gamma^v(Q^2, Q_0^2) V_{35}^{(5)}(Q_0^2) \quad (41)$$

- **T₃, T₈, T₁₅ and T₂₄:**

$$T_{3,8,15,24}^{(5)}(Q^2) = \Gamma^+(Q^2, Q_0^2) T_{3,8,15,24}^{(5)}(Q_0^2) \quad (42)$$

- **T₃₅:**

$$T_{35}^{(5)}(Q^2) = \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, Q_0^2)} \begin{pmatrix} \Sigma^{(5)}(Q_0^2) \\ g^{(5)}(Q_0^2) \end{pmatrix} \quad (43)$$

1.1.4 $m_t^2 < Q_0^2 < Q^2$

- **Singlet and gluon:**

$$\begin{pmatrix} \Sigma^{(6)}(Q^2) \\ g^{(6)}(Q^2) \end{pmatrix} = \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, Q_0^2)} \begin{pmatrix} \Sigma^{(6)}(Q_0^2) \\ g^{(6)}(Q_0^2) \end{pmatrix} \quad (44)$$

- **V:**

$$V^{(6)}(Q^2) = \Gamma^v(Q^2, Q_0^2) V^{(6)}(Q_0^2) \quad (45)$$

- **V₃, V₈, V₁₅, V₂₄ and V₃₅:**

$$V_{3,8,15,24,35}^{(6)}(Q^2) = \Gamma^-(Q^2, Q_0^2) V_{3,8,15,24,35}^{(6)}(Q_0^2) \quad (46)$$

- **T₃, T₈, T₁₅, T₂₄ and T₃₅:**

$$T_{3,8,15,24,35}^{(6)}(Q^2) = \Gamma^+(Q^2, Q_0^2) T_{3,8,15,24,35}^{(6)}(Q_0^2) \quad (47)$$

1.2 Matching Conditions on the Evolution Kernels: 1 Threshold Crossing

In order to implement the matching conditions in our code, we will show how to transfer them from the PDFs to the evolution kernels.

In this Section we suppose that the evolution crosses only one threshold. We will show how the matching conditions on the PDFs modify the form of the evolution kernels in the cases: 1) $Q_0^2 < m_c^2 \leq Q^2$, 2) $Q_0^2 < m_b^2 \leq Q^2$ and $Q_0^2 < m_c^2 \leq Q^2$.

1.2.1 $Q_0^2 < m_c^2 \leq Q^2$

In order to evolve PDFs from the scale Q_0^2 to Q^2 passing through the threshold m_c^2 , we have to: first evolve them from Q_0^2 to m_c^2 , where there are 3 active flavours, then increase the number of active flavour from 3 to 4 by imposing the matching conditions, and in the end evolve the PDFs, now having 4 active flavours, from m_c^2 to the scale Q^2 .

In what follows we will work only in the Mellin space, so we will drop any dependence on N .

Let's start with singlet and gluon. We have:

$$\begin{pmatrix} \Sigma^{(4)} \\ g^{(4)} \end{pmatrix}(Q^2) = \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}(Q^2, m_c^2) \begin{pmatrix} \Sigma^{(4)} \\ g^{(4)} \end{pmatrix}(m_c^2). \quad (48)$$

From eq. (23):

$$\begin{pmatrix} \Sigma^{(4)} \\ g^{(4)} \end{pmatrix}(m_c^2) = \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_s^2(m_c^2) A_{qq,c}^{NS,(2)} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + a_s^2(m_c^2) \begin{pmatrix} \tilde{A}_{cq}^{S,(2)} & \tilde{A}_{cg}^{S,(2)} \\ A_{gq,c}^{S,(2)} & A_{gg,c}^{S,(2)} \end{pmatrix} \right] \begin{pmatrix} \Sigma^{(3)} \\ g^{(3)} \end{pmatrix}(m_c^2) \quad (49)$$

now, substituting the above relation into the eq. (48), we get:

$$\begin{aligned} \begin{pmatrix} \Sigma^{(4)} \\ g^{(4)} \end{pmatrix}(Q^2) &= \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}(Q^2, m_c^2) \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_s^2(m_c^2) A_{qq,c}^{NS,(2)} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \right. \\ &\quad \left. a_s^2(m_c^2) \begin{pmatrix} \tilde{A}_{cq}^{S,(2)} & \tilde{A}_{cg}^{S,(2)} \\ A_{gq,c}^{S,(2)} & A_{gg,c}^{S,(2)} \end{pmatrix} \right] \begin{pmatrix} \Sigma^{(3)} \\ g^{(3)} \end{pmatrix}(m_c^2). \end{aligned} \quad (50)$$

But:

$$\begin{pmatrix} \Sigma^{(3)} \\ g^{(3)} \end{pmatrix}(m_c^2) = \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}(m_c^2, Q_0^2) \begin{pmatrix} \Sigma^{(3)} \\ g^{(3)} \end{pmatrix}(Q_0^2). \quad (51)$$

So, in the end:

$$\begin{aligned} \begin{pmatrix} \Sigma^{(4)} \\ g^{(4)} \end{pmatrix}(Q^2) &= \left\{ \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}(Q^2, m_c^2) \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_s^2(m_c^2) A_{qq,c}^{NS,(2)} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \right. \right. \\ &\quad \left. \left. a_s^2(m_c^2) \begin{pmatrix} \tilde{A}_{cq}^{S,(2)} & \tilde{A}_{cg}^{S,(2)} \\ A_{gq,c}^{S,(2)} & A_{gg,c}^{S,(2)} \end{pmatrix} \right] \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}(m_c^2, Q_0^2) \right\} \begin{pmatrix} \Sigma^{(3)} \\ g^{(3)} \end{pmatrix}(Q_0^2) \end{aligned} \quad (52)$$

Now we consider the distributions V , $V_{3,8}$ and $T_{3,8}$ which evolve respectively through Γ^v , Γ^- and Γ^+ , but which obey the same matching conditions. So:

$$P^{(4)}(Q^2) = \Gamma^{(P)}(Q^2, m_c^2) P^{(4)}(m_c^2) \quad (53)$$

so that:

$$P = \begin{cases} V & \rightarrow \Gamma^{(P)} = \Gamma^v \\ V_{3,8} & \rightarrow \Gamma^{(P)} = \Gamma^- \\ T_{3,8} & \rightarrow \Gamma^{(P)} = \Gamma^+ \end{cases} \quad (54)$$

but:

$$P^{(4)}(m_c^2) = [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] P^{(3)}(m_c^2) \quad (55)$$

and:

$$P^{(3)}(m_c^2) = \Gamma^{(P)}(m_c^2, Q_0^2) P^{(3)}(Q_0^2) \quad (56)$$

so that:

$$P^{(4)}(Q^2) = \left\{ \Gamma^{(P)}(Q^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^{(P)}(m_c^2, Q_0^2) \right\} P^{(3)}(Q_0^2) \quad (57)$$

Now we consider V_{15} , which before m_c^2 evolves as:

$$V_{15}^{(3)}(m_c^2) = \Gamma^v(m_c^2, Q_0^2) V^{(3)}(Q_0^2) \quad (58)$$

while after m_c^2 it evolves as:

$$V_{15}^{(4)}(Q^2) = \Gamma^-(Q_0^2, m_c^2) V_{15}^{(4)}(m_c^2). \quad (59)$$

From eq. (24), we find that the matching condition at m_c^2 is:

$$V_{15}^{(4)}(m_c^2) = [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] V_{15}^{(3)}(m_c^2) \quad (60)$$

so:

$$V_{15}^{(4)}(Q^2) = \left\{ \Gamma^-(Q^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (61)$$

Instead, for $Q_0^2 < m_c^2 \leq Q^2$, both V_{24} and V_{35} evolve as:

$$V_{24,35}^{(4)}(Q^2) = \left\{ \Gamma^v(Q^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (62)$$

Now, we consider T_{15} . After m_c^2 , it evolves as:

$$T_{15}^{(4)}(Q^2) = \Gamma^+(Q^2, m_c^2) T_{15}^{(4)}(m_c^2). \quad (63)$$

This time the matching condition is given by the first line of eq. (18):

$$T_{15}^{(4)}(m_c^2) = \begin{pmatrix} 1 + a_s^2(m_c^2) [A_{qq,c}^{NS,(2)} - 3\tilde{A}_{cq}^{S,(2)}] & -3a_s^2(m_c^2) \tilde{A}_{cg}^{S,(2)} \end{pmatrix} \begin{pmatrix} \Sigma^{(3)}(m_c^2) \\ g^{(3)}(m_c^2) \end{pmatrix}. \quad (64)$$

But:

$$\begin{pmatrix} \Sigma^{(3)}(m_c^2) \\ g^{(3)}(m_c^2) \end{pmatrix} = \begin{pmatrix} \Gamma_{qq}(m_c^2, Q_0^2) & \Gamma_{qg}(m_c^2, Q_0^2) \\ \Gamma_{gq}(m_c^2, Q_0^2) & \Gamma_{gg}(m_c^2, Q_0^2) \end{pmatrix} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (65)$$

In the end, one finds that:

$$\begin{aligned} T_{15}^{(4)}(Q^2) &= \left\{ \Gamma^+(Q^2, m_c^2) \begin{pmatrix} 1 + a_s^2(m_c^2) [A_{qq,c}^{NS,(2)} - 3\tilde{A}_{cq}^{S,(2)}] & -3a_s^2(m_c^2) \tilde{A}_{cg}^{S,(2)} \end{pmatrix} \right. \\ &\quad \left. \begin{pmatrix} \Gamma_{qq}(m_c^2, Q_0^2) & \Gamma_{qg}(m_c^2, Q_0^2) \\ \Gamma_{gq}(m_c^2, Q_0^2) & \Gamma_{gg}(m_c^2, Q_0^2) \end{pmatrix} \right\} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \end{aligned} \quad (66)$$

Now we are left only with T_{24} and T_{35} , which evolve as the single before and after the threshold, so they evolve exactly as the first line of eq. (52), i.e:

$$\begin{aligned} T_{24,35}^{(4)}(Q^2) &= \left\{ \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \end{pmatrix} (Q^2, m_c^2) \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_s^2(m_c^2) A_{qq,c}^{NS,(2)} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \right. \right. \\ &\quad \left. \left. a_s^2(m_c^2) \begin{pmatrix} \tilde{A}_{cq}^{S,(2)} & \tilde{A}_{cg}^{S,(2)} \\ A_{gq,c}^{S,(2)} & A_{gg,c}^{S,(2)} \end{pmatrix} \right] \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix} (m_c^2, Q_0^2) \right\} \begin{pmatrix} \Sigma^{(3)} \\ g^{(3)} \end{pmatrix} (Q_0^2) \end{aligned} \quad (67)$$

Now, let us summarise what happens to the evolution kernels, by introducing the matching conditions, if one crosses the m_c^2 threshold. We remind that the matching conditions appear only from the NNLO.

• **Singlet and gluon:**

$$\begin{pmatrix} \Sigma^{(4)}(Q^2) \\ g^{(4)}(Q^2) \end{pmatrix} = \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix} \begin{pmatrix} M_{11}^c & M_{12}^c \\ M_{21}^c & M_{22}^c \end{pmatrix}}_{(Q^2, m_c^2)} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (68)$$

where:

$$\begin{pmatrix} M_{11}^c & M_{12}^c \\ M_{21}^c & M_{22}^c \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_s^2(m_c^2) A_{qq,c}^{NS,(2)} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + a_s^2(m_c^2) \begin{pmatrix} \tilde{A}_{cq}^{S,(2)} & \tilde{A}_{cg}^{S,(2)} \\ A_{gq,c}^{S,(2)} & A_{gg,c}^{S,(2)} \end{pmatrix} \quad (69)$$

• **V:**

$$V^{(4)}(Q^2) = \left\{ \Gamma^v(Q^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (70)$$

• **V₃ and V₈:**

$$V_{3,8}^{(4)}(Q^2) = \left\{ \Gamma^-(Q^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^-(m_c^2, Q_0^2) \right\} V_{3,8}^{(3)}(Q_0^2) \quad (71)$$

• **V₁₅:**

$$V_{15}^{(4)}(Q^2) = \left\{ \Gamma^-(Q^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (72)$$

• **V₂₄ and V₃₅:**

$$V_{24,35}^{(4)}(Q^2) = \left\{ \Gamma^v(Q^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (73)$$

• **T₃ and T₈:**

$$T_{3,8}^{(4)}(Q^2) = \left\{ \Gamma^+(Q^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^+(m_c^2, Q_0^2) \right\} T_{3,8}^{(3)}(Q_0^2) \quad (74)$$

• **T₁₅:**

$$\begin{aligned} T_{15}^{(4)}(Q^2) &= \left\{ \Gamma^+(Q^2, m_c^2) \left(1 + a_s^2(m_c^2) [A_{qq,c}^{NS,(2)} - 3\tilde{A}_{cq}^{S,(2)}] \right) - 3a_s^2(m_c^2) \tilde{A}_{cg}^{S,(2)} \right\} \times \\ &\quad \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \left\{ \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \right\} \end{aligned} \quad (75)$$

• **T₂₄ and T₃₅:**

$$T_{24,35}^{(4)}(Q^2) = \left\{ \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \end{pmatrix}}_{(Q^2, m_c^2)} \begin{pmatrix} M_{11}^c & M_{12}^c \\ M_{21}^c & M_{22}^c \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (76)$$

Now, it is very easy to rewrite the above summary for the crossing of the remaining thresholds m_b^2 ($Q_0^2 < m_b^2 \leq Q^2$) and m_t^2 ($Q_0^2 < m_t^2 \leq Q^2$).

1.2.2 $Q_0^2 < m_b^2 \leq Q^2$

• **Singlet and gluon:**

$$\begin{pmatrix} \Sigma^{(5)}(Q^2) \\ g^{(5)}(Q^2) \end{pmatrix} = \left\{ \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, m_b^2)} \begin{pmatrix} M_{11}^b & M_{12}^b \\ M_{21}^b & M_{22}^b \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(4)}(Q_0^2) \\ g^{(4)}(Q_0^2) \end{pmatrix} \quad (77)$$

where:

$$\begin{pmatrix} M_{11}^b & M_{12}^b \\ M_{21}^b & M_{22}^b \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_s^2(m_b^2) A_{qq,b}^{NS,(2)} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + a_s^2(m_b^2) \begin{pmatrix} \tilde{A}_{bq}^{S,(2)} & \tilde{A}_{bg}^{S,(2)} \\ A_{gq,b}^{S,(2)} & A_{gg,b}^{S,(2)} \end{pmatrix} \quad (78)$$

- **V:**

$$V^{(5)}(Q^2) = \left\{ \Gamma^v(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^v(m_b^2, Q_0^2) \right\} V^{(4)}(Q_0^2) \quad (79)$$

- **V₃, V₈ and V₁₅:**

$$V_{3,8,15}^{(5)}(Q^2) = \left\{ \Gamma^-(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^-(m_b^2, Q_0^2) \right\} V_{3,8,15}^{(4)}(Q_0^2) \quad (80)$$

- **V₂₄:**

$$V_{24}^{(5)}(Q^2) = \left\{ \Gamma^-(Q_0^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^v(m_b^2, Q_0^2) \right\} V^{(4)}(Q_0^2) \quad (81)$$

- **V₃₅:**

$$V_{35}^{(5)}(Q^2) = \left\{ \Gamma^v(Q_0^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^v(m_b^2, Q_0^2) \right\} V^{(4)}(Q_0^2) \quad (82)$$

- **T₃, T₈ and T₁₅:**

$$T_{3,8,15}^{(5)}(Q^2) = \left\{ \Gamma^+(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^+(m_b^2, Q_0^2) \right\} T_{3,8,15}^{(4)}(Q_0^2) \quad (83)$$

- **T₂₄:**

$$T_{24}^{(5)}(Q^2) = \left\{ \Gamma^+(Q^2, m_b^2) \left(1 + a_s^2(m_b^2) [A_{qq,b}^{NS,(2)} - 4\tilde{A}_{bq}^{S,(2)}] \right) - 4a_s^2(m_b^2) \tilde{A}_{bg}^{S,(2)} \right\} \times$$

$$\underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, Q_0^2)} \left\{ \begin{pmatrix} \Sigma^{(4)}(Q_0^2) \\ g^{(4)}(Q_0^2) \end{pmatrix} \right\} \quad (84)$$

- **T₃₅:**

$$T_{35}^{(5)}(Q^2) = \left\{ \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, m_b^2)} \begin{pmatrix} M_{11}^b & M_{12}^b \\ M_{21}^b & M_{22}^b \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(4)}(Q_0^2) \\ g^{(4)}(Q_0^2) \end{pmatrix} \quad (85)$$

1.2.3 $Q_0^2 < m_t^2 \leq Q^2$

- **Singlet and gluon:**

$$\begin{pmatrix} \Sigma^{(6)}(Q^2) \\ g^{(6)}(Q^2) \end{pmatrix} = \left\{ \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, m_t^2)} \begin{pmatrix} M_{11}^t & M_{12}^t \\ M_{21}^t & M_{22}^t \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_t^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(5)}(Q_0^2) \\ g^{(5)}(Q_0^2) \end{pmatrix} \quad (86)$$

where:

$$\begin{pmatrix} M_{11}^t & M_{12}^t \\ M_{21}^t & M_{22}^t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_s^2(m_t^2) A_{qq,t}^{NS,(2)} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + a_s^2(m_t^2) \begin{pmatrix} \tilde{A}_{tq}^{S,(2)} & \tilde{A}_{tg}^{S,(2)} \\ A_{gq,t}^{S,(2)} & A_{gg,t}^{S,(2)} \end{pmatrix} \quad (87)$$

- **V:**

$$V^{(6)}(Q^2) = \left\{ \Gamma^v(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \Gamma^v(m_t^2, Q_0^2) \right\} V^{(5)}(Q_0^2) \quad (88)$$

- **V₃, V₈, V₁₅ and V₂₄:**

$$V_{3,8,15,24}^{(6)}(Q^2) = \left\{ \Gamma^-(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \Gamma^-(m_t^2, Q_0^2) \right\} V_{3,8,15,24}^{(5)}(Q_0^2) \quad (89)$$

- **V₃₅:**

$$V_{35}^{(6)}(Q^2) = \left\{ \Gamma^-(Q_0^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \Gamma^v(m_t^2, Q_0^2) \right\} V^{(5)}(Q_0^2) \quad (90)$$

- **T₃, T₈, T₁₅ and T₂₄:**

$$T_{3,8,15,24}^{(6)}(Q^2) = \left\{ \Gamma^+(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \Gamma^+(m_t^2, Q_0^2) \right\} T_{3,8,15,24}^{(5)}(Q_0^2) \quad (91)$$

- **T₃₅:**

$$T_{35}^{(6)}(Q^2) = \left\{ \Gamma^+(Q^2, m_t^2) \left(1 + a_s^2(m_t^2) [A_{qq,t}^{NS,(2)} - 5\tilde{A}_{tq}^{S,(2)}] - 5a_s^2(m_t^2) \tilde{A}_{tg}^{S,(2)} \right) \right. \\ \left. \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_t^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(5)}(Q_0^2) \\ g^{(5)}(Q_0^2) \end{pmatrix} \quad (92)$$

1.3 Matching Conditions on the Evolution Kernels: 2 Thresholds Crossing

In this Section we will discuss the case in which the evolution crosses two thresholds. Therefore there are only two situations: 1) $Q_0^2 < m_c^2 < m_b^2 \leq Q^2$ and 2) $Q_0^2 < m_b^2 < m_t^2 \leq Q^2$. Anyway, there is nothing new, indeed to obtain such evolution kernels we have just to “merge” together what we have already done in the previous Section.

1.3.1 $Q_0^2 < m_c^2 < m_b^2 \leq Q^2$

- **Singlet and gluon:**

$$\begin{pmatrix} \Sigma^{(5)}(Q^2) \\ g^{(5)}(Q^2) \end{pmatrix} = \left\{ \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, m_b^2)} \begin{pmatrix} M_{11}^b & M_{12}^b \\ M_{21}^b & M_{22}^b \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, m_c^2)} \right. \\ \left. \underbrace{\begin{pmatrix} M_{11}^c & M_{12}^c \\ M_{21}^c & M_{22}^c \end{pmatrix}}_{(m_c^2, Q_0^2)} \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix} \right\} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (93)$$

- **V:**

$$V^{(5)}(Q^2) = \left\{ \Gamma^v(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \times \right. \\ \left. \Gamma^v(m_b^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (94)$$

- **V₃ and V₈:**

$$V_{3,8}^{(5)}(Q^2) = \left\{ \Gamma^-(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \times \right. \\ \left. \Gamma^-(m_b^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^-(m_c^2, Q_0^2) \right\} V_{3,8}^{(3)}(Q_0^2) \quad (95)$$

- **V₁₅:**

$$V_{15}^{(5)}(Q^2) = \left\{ \Gamma^-(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \times \right. \\ \left. \Gamma^-(m_b^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (96)$$

- **V₂₄**:

$$V_{24}^{(5)}(Q^2) = \left\{ \Gamma^-(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \times \right. \\ \left. \Gamma^v(m_b^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (97)$$

- **V₃₅**:

$$V_{35}^{(5)}(Q^2) = \left\{ \Gamma^v(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \times \right. \\ \left. \Gamma^v(m_b^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (98)$$

- **T₃ and T₈**:

$$T_{3,8}^{(5)}(Q^2) = \left\{ \Gamma^+(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \times \right. \\ \left. \Gamma^+(m_b^2, m_c^2) [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^+(m_c^2, Q_0^2) \right\} T_{3,8}^{(3)}(Q_0^2) \quad (99)$$

- **T₁₅**:

$$T_{15}^{(5)}(Q^2) = \left\{ \Gamma^+(Q^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^+(m_b^2, m_c^2) \times \right. \\ \left. \left(1 + a_s^2(m_c^2) [A_{qq,c}^{NS,(2)} - 3\tilde{A}_{cq}^{S,(2)}] \quad -3a_s^2(m_c^2) \tilde{A}_{cg}^{S,(2)} \right) \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (100)$$

- **T₂₄**:

$$T_{24}^{(5)}(Q^2) = \left\{ \Gamma^+(Q^2, m_b^2) \left(1 + a_s^2(m_b^2) [A_{qq,b}^{NS,(2)} - 4\tilde{A}_{bq}^{S,(2)}] \quad -4a_s^2(m_b^2) \tilde{A}_{bg}^{S,(2)} \right) \times \right. \\ \left. \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, m_c^2)} \begin{pmatrix} M_{11}^c & M_{12}^c \\ M_{21}^c & M_{22}^c \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (101)$$

- **T₃₅**:

$$T_{35}^{(5)}(Q^2) = \left\{ \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, m_b^2)} \begin{pmatrix} M_{11}^b & M_{12}^b \\ M_{21}^b & M_{22}^b \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, m_c^2)} \times \right. \\ \left. \begin{pmatrix} M_{11}^c & M_{12}^c \\ M_{21}^c & M_{22}^c \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (102)$$

1.3.2 $Q_0^2 < m_b^2 < m_t^2 \leq Q^2$

- **Singlet and gluon:**

$$\begin{aligned} \begin{pmatrix} \Sigma^{(6)}(Q^2) \\ g^{(6)}(Q^2) \end{pmatrix} &= \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix} \begin{pmatrix} M_{11}^t & M_{12}^t \\ M_{21}^t & M_{22}^t \end{pmatrix}}_{(Q^2, m_t^2)} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_t^2, m_b^2)} \times \\ &\quad \underbrace{\begin{pmatrix} M_{11}^b & M_{12}^b \\ M_{21}^b & M_{22}^b \end{pmatrix} \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, Q_0^2)} \begin{pmatrix} \Sigma^{(4)}(Q_0^2) \\ g^{(4)}(Q_0^2) \end{pmatrix} \end{aligned} \quad (103)$$

- **V:**

$$\begin{aligned} V^{(6)}(Q^2) &= \left\{ \Gamma^v(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\ &\quad \left. \Gamma^v(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^v(m_b^2, Q_0^2) \right\} V^{(4)}(Q_0^2) \end{aligned} \quad (104)$$

- **V₃, V₈ and V₁₅:**

$$\begin{aligned} V_{3,8,15}^{(6)}(Q^2) &= \left\{ \Gamma^-(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\ &\quad \left. \Gamma^-(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^-(m_b^2, Q_0^2) \right\} V_{3,8,15}^{(4)}(Q_0^2) \end{aligned} \quad (105)$$

- **V₂₄:**

$$\begin{aligned} V_{24}^{(6)}(Q^2) &= \left\{ \Gamma^-(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\ &\quad \left. \Gamma^-(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^v(m_b^2, Q_0^2) \right\} V^{(4)}(Q_0^2) \end{aligned} \quad (106)$$

- **V₃₅:**

$$\begin{aligned} V_{35}^{(6)}(Q^2) &= \left\{ \Gamma^-(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\ &\quad \left. \Gamma^v(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^v(m_b^2, Q_0^2) \right\} V^{(4)}(Q_0^2) \end{aligned} \quad (107)$$

- **T₃, T₈ and T₁₅:**

$$\begin{aligned} T_{3,8,15}^{(6)}(Q^2) &= \left\{ \Gamma^+(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\ &\quad \left. \Gamma^+(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^+(m_b^2, Q_0^2) \right\} T_{3,8,15}^{(4)}(Q_0^2) \end{aligned} \quad (108)$$

- **T₂₄:**

$$\begin{aligned} T_{24}^{(6)}(Q^2) &= \left\{ \Gamma^+(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \Gamma^+(m_t^2, m_b^2) \times \right. \\ &\quad \left. \underbrace{\begin{pmatrix} 1 + a_s^2(m_b^2) [A_{qq,b}^{NS,(2)} - 4\tilde{A}_{bq}^{S,(2)}] & -4a_s^2(m_b^2) \tilde{A}_{bg}^{S,(2)} \end{pmatrix} \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(4)}(Q_0^2) \\ g^{(4)}(Q_0^2) \end{pmatrix} \end{aligned} \quad (109)$$

- **T₃₅:**

$$T_{35}^{(6)}(Q^2) = \left\{ \Gamma^+(Q^2, m_t^2) \left(1 + a_s^2(m_t^2) [A_{qq,t}^{NS,(2)} - 5\tilde{A}_{tq}^{S,(2)}] - 5a_s^2(m_t^2) \tilde{A}_{tg}^{S,(2)} \right) \times \right. \\ \left. \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_t^2, m_b^2)} \begin{pmatrix} M_{11}^b & M_{12}^b \\ M_{21}^b & M_{22}^b \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(4)}(Q_0^2) \\ g^{(4)}(Q_0^2) \end{pmatrix} \quad (110)$$

1.4 Matching Conditions on the Evolution Kernels: 3 Thresholds Crossing

In this Section we will discuss the case in which the evolution crosses three thresholds. Therefore there is only one situation: $Q_0^2 < m_c^2 < m_b^2, m_t^2 \leq Q^2$

1.4.1 $Q_0^2 < m_c^2 < m_b^2 < m_t^2 \leq Q^2$

- **Singlet and gluon:**

$$\begin{pmatrix} \Sigma^{(6)}(Q^2) \\ g^{(6)}(Q^2) \end{pmatrix} = \left\{ \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(Q^2, m_t^2)} \begin{pmatrix} M_{11}^t & M_{12}^t \\ M_{21}^t & M_{22}^t \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_t^2, m_b^2)} \times \right. \\ \left. \begin{pmatrix} M_{11}^b & M_{12}^b \\ M_{21}^b & M_{22}^b \end{pmatrix} \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, m_c^2)} \begin{pmatrix} M_{11}^c & M_{12}^c \\ M_{21}^c & M_{22}^c \end{pmatrix} \right. \\ \left. \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \quad (111)$$

- **V:**

$$V^{(6)}(Q^2) = \left\{ \Gamma^v(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\ \Gamma^v(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^v(m_b^2, m_c^2) \\ \left. [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (112)$$

- **V₃ and V₈:**

$$V_{3,8}^{(6)}(Q^2) = \left\{ \Gamma^-(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\ \Gamma^-(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^-(m_b^2, m_c^2) \\ \left. [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^-(m_c^2, Q_0^2) \right\} V_{3,8}^{(3)}(Q_0^2) \quad (113)$$

- **V₁₅:**

$$V_{15}^{(6)}(Q^2) = \left\{ \Gamma^-(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\ \Gamma^-(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^-(m_b^2, m_c^2) \\ \left. [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2) \quad (114)$$

- **V₂₄**:

$$\begin{aligned}
V_{15}^{(6)}(Q^2) &= \left\{ \Gamma^-(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\
&\quad \Gamma^-(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^v(m_b^2, m_c^2) \\
&\quad \left. [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2)
\end{aligned} \tag{115}$$

- **V₃₅**:

$$\begin{aligned}
V_{15}^{(6)}(Q^2) &= \left\{ \Gamma^-(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\
&\quad \Gamma^v(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^v(m_b^2, m_c^2) \\
&\quad \left. [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^v(m_c^2, Q_0^2) \right\} V^{(3)}(Q_0^2)
\end{aligned} \tag{116}$$

- **T₃** and **T₈**:

$$\begin{aligned}
T_{3,8}^{(6)}(Q^2) &= \left\{ \Gamma^+(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\
&\quad \Gamma^+(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^+(m_b^2, m_c^2) \\
&\quad \left. [1 + a_s^2(m_c^2) A_{qq,c}^{NS,(2)}] \Gamma^+(m_c^2, Q_0^2) \right\} T_{3,8}^{(3)}(Q_0^2)
\end{aligned} \tag{117}$$

- **T₁₅**:

$$\begin{aligned}
T_{15}^{(6)}(Q^2) &= \left\{ \Gamma^+(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \times \right. \\
&\quad \Gamma^+(m_t^2, m_b^2) [1 + a_s^2(m_b^2) A_{qq,b}^{NS,(2)}] \Gamma^+(m_b^2, m_c^2) \\
&\quad \left. \left(1 + a_s^2(m_c^2) [A_{qq,c}^{NS,(2)} - 3\tilde{A}_{cq}^{S,(2)}] - 3a_s^2(m_c^2) \tilde{A}_{cg}^{S,(2)} \right) \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix}
\end{aligned} \tag{118}$$

- **T₂₄**:

$$\begin{aligned}
T_{24}^{(6)}(Q^2) &= \left\{ \Gamma^+(Q^2, m_t^2) [1 + a_s^2(m_t^2) A_{qq,t}^{NS,(2)}] \Gamma^+(m_t^2, m_b^2) \times \right. \\
&\quad \left. \left(1 + a_s^2(m_b^2) [A_{qq,b}^{NS,(2)} - 4\tilde{A}_{bq}^{S,(2)}] - 4a_s^2(m_b^2) \tilde{A}_{bg}^{S,(2)} \right) \underbrace{\begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, m_c^2)} \right\} \\
&\quad \underbrace{\begin{pmatrix} M_{11}^c & M_{12}^c \\ M_{21}^c & M_{22}^c \end{pmatrix} \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \left\{ \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix} \right\}
\end{aligned} \tag{119}$$

• **T₃₅**:

$$\begin{aligned}
T_{35}^{(6)}(Q^2) &= \left\{ \Gamma^+(Q^2, m_t^2) \left(1 + a_s^2(m_b^2) [A_{qq,t}^{NS,(2)} - 5\tilde{A}_{tq}^{S,(2)}] - 5a_s^2(m_t^2) \tilde{A}_{tg}^{S,(2)} \right) \right. \\
&\quad \left. \underbrace{\begin{pmatrix} M_{11}^b & M_{12}^b \\ M_{21}^b & M_{22}^b \end{pmatrix} \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_b^2, m_c^2)} \right. \\
&\quad \left. \underbrace{\begin{pmatrix} M_{11}^c & M_{12}^c \\ M_{21}^c & M_{22}^c \end{pmatrix} \begin{pmatrix} \Gamma_{qq} & \Gamma_{qg} \\ \Gamma_{gq} & \Gamma_{gg} \end{pmatrix}}_{(m_c^2, Q_0^2)} \right\} \begin{pmatrix} \Sigma^{(3)}(Q_0^2) \\ g^{(3)}(Q_0^2) \end{pmatrix}
\end{aligned} \tag{120}$$

2 Evolution and matching with intrinsic distributions

The (quite outdated) discussion presented above concerning the PDF matching conditions is restricted to forward evolution (*i.e.* $Q > Q_0$) and under the assumption that possible intrinsic heavy-quark contributions are absent.² This corresponds to assuming that heavy-quark PDFs are entirely dynamically generated at the respective threshold and are identically vanishing below threshold. While this turns out to be a good approximation for PDFs, this is not the case for fragmentation functions (FFs). Indeed, intrinsic heavy-quark FFs are sizeable. In addition, assuming that there are no heavy-quark PDFs makes backward evolution problematic. The basic problem is that the matching-function matrix is non-squared which forbids its inversion thus complicating backward evolution.

The purpose of this section is the relaxation of the no-intrinsic-PDF assumption. This will imply the introduction of new matching functions due to the presence of heavy-quark PDFs below threshold. Importantly, the perturbative expansion of these additional functions is different from zero at $\mathcal{O}(\alpha_s)$ even choosing $\mu_h = m_h$, with μ_h the matching scale, producing a discontinuity already at NLO. However, in what follows we will assume that matching conditions are not computed beyond $\mathcal{O}(\alpha_s^2)$, *i.e.* NNLO, and that intrinsic heavy-quark and heavy-antiquark contributions are equal below threshold. These assumptions lead to:

$$h^{(n_f)}(\mu_h) = \bar{h}^{(n_f)}(\mu_h) \quad \text{and} \quad h^{(n_f+1)}(\mu_h) = \bar{h}^{(n_f+1)}(\mu_h), \tag{121}$$

where we dropped the non-scale dependence. It immediately follows that the valence distributions $\{V, V_3, V_8, V_{15}, V_{24}, V_{35}\}$ match multiplicatively as:³

$$V_i^{(n_f+1)}(\mu_h) = [1 + K_{il}](\mu_h) V_i^{(n_f)}(\mu_h) = [1 + K_{il}(\mu_h)] V_i^{(n_f)}(\mu_h), \quad i = 3, 8, 15, 24, 35, \tag{122}$$

with K_{il} some given function that starts at $\mathcal{O}(\alpha_s^2)$. This equivalently amounts to saying that no non-perturbative quark-antiquark asymmetry is originally present nor is generated by the matching procedure.⁴

This allows us to concentrate the discussion on the singlet sector. We start by generalising Eqs. (1),

²In fact, the discussion also assumes that the PDF matching takes place at the heavy quark-mass value. This is also a non-necessary assumption that we will however not discuss here.

³Note that “1” in Eq. (122) is to be understood as $\delta(1-x)$ in x space.

⁴The assumption of absence of non-perturbative quark-antiquark asymmetry is badly violated for light-valence distributions, such as up and down in the proton. However, we will never realistically need to match PDFs or FFs at the light-quark scales. On the other hand, this assumption helps us keep the discussion general.

(2), and (3) (changing and simplifying a bit the notation) as:

$$\begin{aligned}
g^{(n_f+1)} &= [1 + K_{gg}]g^{(n_f)} + K_{gl} \sum_l l^{+(n_f)} + K_{gh}h^{+(n_f)}, \\
l^{+(n_f+1)} &= [1 + K_{ll}]l^{+(n_f)}, \\
h^{+(n_f+1)} &= K_{hg}g^{(n_f)} + K_{hl} \sum_l l^{+(n_f)} + [1 + K_{hh}]h^{+(n_f)}, \\
H^{+(n_f+1)} &= H^{+(n_f)},
\end{aligned} \tag{123}$$

where $q^+ = q + \bar{q}$ and l runs between 1 and n_f . We remind that h is the $(n_f + 1)$ -th heavier flavour, *i.e.* the one that becomes active at the threshold under consideration. We have also introduced the “super” heavy quark flavour H , heavier than h , that is not affected by the matching at the n_f -th threshold. Note that there are two new functions coming into play, namely K_{gh} and K_{hh} , both multiplying the intrinsic heavy-quark contribution $h^{+(n_f)}$. Moreover, we have written Eq. (123) in such a way that the coefficients K start at least at $\mathcal{O}(\alpha_s)$. In fact, we have already seen that $K_{ll} = \mathcal{O}(\alpha_s^2)$; the same applies to K_{hl} and K_{gl} . Defining the column vector P with components $(g, d^+, u^+, s^+, c^+, b^+, t^+)$, we can represent the matching in a matrix form as:

$$P^{(n_f+1)} = [\mathbb{I} + \mathbb{K}^{(n_f)}] P^{(n_f)}, \tag{124}$$

where $\mathbb{I}$ is the 7×7 unity matrix and $\mathbb{K}^{(n_f)}$ are given, depending on the value of n_f , by:

$$\begin{aligned}
\mathbb{K}^{(0)} &= \begin{pmatrix} K_{gg} & K_{gh} & 0 & 0 & 0 & 0 & 0 \\ K_{hg} & K_{hh} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} & \mathbb{K}^{(1)} &= \begin{pmatrix} K_{gg} & K_{gl} & K_{gh} & 0 & 0 & 0 & 0 \\ 0 & K_{ll} & 0 & 0 & 0 & 0 & 0 \\ K_{hg} & K_{hl} & K_{hh} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\
\mathbb{K}^{(2)} &= \begin{pmatrix} K_{gg} & K_{gl} & K_{gl} & K_{gh} & 0 & 0 & 0 \\ 0 & K_{ll} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & K_{ll} & 0 & 0 & 0 & 0 \\ K_{hg} & K_{hl} & K_{hl} & K_{hh} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} & \mathbb{K}^{(3)} &= \begin{pmatrix} K_{gg} & K_{gl} & K_{gl} & K_{gl} & K_{gh} & 0 & 0 \\ 0 & K_{ll} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & K_{ll} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & K_{ll} & 0 & 0 & 0 \\ K_{hg} & K_{hl} & K_{hl} & K_{hl} & K_{hh} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\
\mathbb{K}^{(4)} &= \begin{pmatrix} K_{gg} & K_{gl} & K_{gl} & K_{gl} & K_{gl} & K_{gh} & 0 \\ 0 & K_{ll} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & K_{ll} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & K_{ll} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & K_{ll} & 0 & 0 \\ K_{hg} & K_{hl} & K_{hl} & K_{hl} & K_{hl} & K_{hh} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} & \mathbb{K}^{(5)} &= \begin{pmatrix} K_{gg} & K_{gl} & K_{gl} & K_{gl} & K_{gl} & K_{gl} & K_{gh} \\ 0 & K_{ll} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & K_{ll} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & K_{ll} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & K_{ll} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & K_{ll} & 0 \\ K_{hg} & K_{hl} & K_{hl} & K_{hl} & K_{hl} & K_{hl} & K_{hh} \end{pmatrix}
\end{aligned} \tag{125}$$

The algorithm to determine the matrix $\mathbb{K}^{(n_f)}$, indexing the gluon distribution with 0 and the quark

distributions from 1 to 6, reads:

$$\mathbb{K}_{ij}^{(n_f)} = \begin{cases} K_{gg} & i = j = 0 \\ K_{gl} & i = 0, 1 \leq j \leq n_f \\ K_{gh} & i = 0, j = n_f + 1 \\ K_{ll} & i = j, 1 \leq i, j \leq n_f \\ K_{hg} & i = n_f + 1, j = 0 \\ K_{hl} & i = n_f + 1, 1 \leq j \leq n_f \\ K_{hh} & i = j = n_f + 1 \\ 0 & \text{elsewhere} \end{cases}. \quad (126)$$

Now, in order to apply the matching condition to the distributions in the evolution basis $E = (g, \Sigma, -T_3, T_8, T_{15}, T_{24}, T_{35})$ (notice the minus sign in front of T_3), it is necessary to rotate the matching matrices from the physical basis to the evolution basis through $T\mathbb{K}^{(n_f)}T^{-1}$, where T transforms the vector P into the evolution-basis vector E :

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & -2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & -3 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & -4 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & -5 \end{pmatrix}. \quad (127)$$

The resulting matching matrices are much more complicated and generally less sparse than those in Eq. (125). However, we can still compute them algorithmically depending on the number of light flavours n_f . To do so, we define the following linearly independent combinations:

$$\begin{aligned} M_1 &= K_{gg}, \\ M_2 &= K_{gh} + n_f K_{gl}, \\ M_3 &= K_{gh} - K_{gl}, \\ M_4 &= K_{hg}, \\ M_5 &= K_{hh} + n_f (K_{hl} + K_{ll}), \\ M_6 &= K_{hh} - (K_{hl} + K_{ll}), \\ M_7 &= K_{ll}, \end{aligned} \quad (128)$$

such that:

$$\left[T\mathbb{K}^{(n_f)}T^{-1} \right]_{ij} = \begin{cases} \begin{array}{lll} M_1 & i=0 & j=0 \\ \frac{1}{6}M_2 & i=0 & j=1 \\ 0 & i=0 & 2 \leq j \leq n_f \\ -\frac{1}{n_f+1}M_3 & i=0 & j=n_f+1 \\ \frac{1}{j(j-1)}M_2 & i=0 & n_f+2 \leq j \leq 6 \end{array} \\ \\ \begin{array}{lll} M_4 & i=1 & j=0 \\ \frac{1}{6}M_5 & i=1 & j=1 \\ 0 & i=1 & 2 \leq j \leq n_f \\ -\frac{1}{n_f+1}M_6 & i=1 & j=n_f+1 \\ \frac{1}{j(j-1)}M_5 & i=1 & n_f+2 \leq j \leq 6 \end{array} \\ \\ \delta_{ij}M_7 & 2 \leq i \leq n_f & 0 \leq j \leq 6 \\ \\ \begin{array}{lll} -n_fM_4 & i=n_f+1 & j=0 \\ \frac{n_f}{6}(-M_5 + (n_f+1)M_7) & i=n_f+1 & j=1 \\ 0 & i=n_f+1 & 2 \leq j \leq n_f \\ \frac{1}{n_f+1}(n_fM_6 + (n_f+1)M_7) & i=n_f+1 & j=n_f+1 \\ \frac{n_f}{j(j-1)}(-M_5 + (n_f+1)M_7) & i=n_f+1 & n_f+2 \leq j \leq 6 \end{array} \\ \\ \begin{array}{lll} M_4 & n_f+2 \leq i \leq 6 & j=0 \\ \frac{1}{6}M_5 & n_f+2 \leq i \leq 6 & j=1 \\ 0 & n_f+2 \leq i \leq 6 & 2 \leq j \leq n_f \\ -\frac{1}{n_f+1}M_6 & n_f+2 \leq i \leq 6 & j=n_f+1 \\ \frac{1}{j(j-1)}M_5 & n_f+2 \leq i \leq 6 & n_f+2 \leq j \leq 6 \end{array} \end{cases} \quad (129)$$

In order to be able to perform the backward evolution, it is necessary to invert the matching matrix $\mathbb{I} + \mathbb{K}^{(n_f)}$. This is conveniently done perturbatively by expanding the matrix $[\mathbb{I} + \mathbb{K}^{(n_f)}]^{-1}$ in powers of α_s and truncating the expansion at the appropriate order. Given the assumption discussed above, we are not interested in going beyond $\mathcal{O}(\alpha_s^2)$. In addition, we know that:

$$\mathbb{K}^{(n_f)} = a_s \mathbb{K}^{(1)(n_f)} + a_s^2 \mathbb{K}^{(2)(n_f)} + \mathcal{O}(\alpha_s^3), \quad (130)$$

so that:

$$[\mathbb{I} + \mathbb{K}^{(n_f)}]^{-1} = 1 - a_s \mathbb{K}^{(1)(n_f)} - a_s^2 \left[\mathbb{K}^{(2)(n_f)} - \left(\mathbb{K}^{(1)(n_f)} \right)^2 \right] + \mathcal{O}(\alpha_s^3). \quad (131)$$

But the matrix $\mathbb{K}^{(1)(n_f)}$ is particularly simple because at $\mathcal{O}(\alpha_s)$ the matching functions K_{ll} , K_{hl} , and K_{gl} vanish. Therefore, using Eq. (126) we find:

$$\mathbb{K}_{ij}^{(1)(n_f)} = \delta_{i,0} \left(\delta_{j,0} K_{gg}^{(1)} + \delta_{j,n_f+1} K_{gh}^{(1)} \right) + \delta_{i,n_f+1} \left(\delta_{j,0} K_{hg}^{(1)} + \delta_{j,n_f+1} K_{hh}^{(1)} \right), \quad (132)$$

such that its square has the same structure and can be written as:

$$\begin{aligned} \left(\mathbb{K}^{(1)(n_f)} \right)_{ij}^2 &= \delta_{i,0} \left(\delta_{j,0} \sum_{k=g,h} K_{gk}^{(1)} K_{kg}^{(1)} + \delta_{j,n_f+1} \sum_{k=g,h} K_{gk}^{(1)} K_{kh}^{(1)} \right) \\ &+ \delta_{i,n_f+1} \left(\delta_{j,0} \sum_{k=g,h} K_{hk}^{(1)} K_{kg}^{(1)} + \delta_{j,n_f+1} \sum_{k=g,h} K_{hk}^{(1)} K_{kh}^{(1)} \right), \end{aligned} \quad (133)$$

where, in x space, the product of two K functions is to be understood as a convolution. We report below the expression in x space of the functions $K_{gh}^{(1)}$ and $K_{hh}^{(1)}$ for the matching of PDFs:

$$\begin{aligned} K_{gh}^{(1)}(x) &= 2C_F \frac{1+(1-x)^2}{x} \left(\ln \frac{\mu_h^2}{m_h^2} - 1 - 2 \ln x \right), \\ K_{hh}^{(1)}(x) &= 2C_F \left[\frac{1+x^2}{1-x} \left(\ln \frac{\mu_h^2}{m_h^2} - 1 - 2 \ln(1-x) \right) \right]_+. \end{aligned} \quad (134)$$

We notice that the $\mathcal{O}(\alpha_s^2)$ correction to these functions is currently unknown, therefore we only use the $\mathcal{O}(\alpha_s)$ expressions. For the sake of completeness, we also report here the remaining $\mathcal{O}(\alpha_s)$ matching functions:

$$\begin{aligned} K_{gg}^{(1)}(x) &= -\frac{4}{3}T_R\delta(1-x)\ln\frac{\mu_h^2}{m_h^2}, \\ K_{hg}^{(1)}(x) &= 4T_R[x^2+(1-x)^2]\ln\frac{\mu_h^2}{m_h^2}, \end{aligned} \quad (135)$$

Of course, the matrix $[\mathbb{I} + \mathbb{K}^{(n_f)}]^{-1}$ in Eq. (131) has to be rotated into the evolution basis by means of the transformation matrix T in Eq. (127)

In order to implement the possibility to compute the evolution operator in the VFNS, we need to know how the evolution operator for a given number of active flavours n_f matches at the following threshold. The first step is to generalise the structure of the DGLAP evolution equations in the QCD evolution basis allowing for the presence of (scale-independent) intrinsic contributions. It is important to realise that the DGLAP equations govern the evolution in μ of the collinear distributions but do not give us any information on the non-dynamic part of these distributions, *i.e.* its intrinsic contribution. The starting point is the DGLAP equation for the quark distribution q_k^+ ⁵ in its basic form:

$$\frac{dq_k^+}{d\ln\mu^2} = \frac{dq_k^+}{dt} = 2P_{qg} \otimes g + (P_{qq}^V + P_{q\bar{q}}^V) \otimes q_k^+ + (P_{qq}^S + P_{q\bar{q}}^S) \otimes \sum_{i=1}^{n_f} q_i^+. \quad (136)$$

It is crucial to notice that the sum on the r.h.s. of the equation above runs over the n_f active flavours only. Now, we sum the index k up to n_f and we get:

$$\frac{d}{dt} \sum_{k=1}^{n_f} q_k^+ = 2n_f P_{qg} \otimes g + P_{qq} \otimes \sum_{k=1}^{n_f} q_k^+, \quad (137)$$

where we have defined:

$$P_{qq} = (P_{qq}^V + P_{q\bar{q}}^V) + n_f(P_{qq}^S + P_{q\bar{q}}^S). \quad (138)$$

In the presence of intrinsic heavy-quark distributions and after some algebra, one finds that:

$$\frac{d\Sigma}{dt} = 2n_f P_{qg} \otimes g + n_f P_{qq} \otimes \left(\frac{\Sigma}{6} + \sum_{j=n_f+1}^6 \frac{E_j}{j(j-1)} \right), \quad (139)$$

with $E_j \in \{\Sigma, -T_3, T_8, T_{15}, T_{24}, T_{35}\}$. Therefore, the singlet distribution, on top of the gluon, couples also to the other quark distributions. Notice that:

$$\sum_{j=n_f+1}^6 \frac{1}{j(j-1)} = \frac{1}{n_f} - \frac{1}{6}, \quad (140)$$

such that, for $E_j = \Sigma$, Eq. (139) reproduces the usual form of the DGLAP equation for the singlet. We now need to determine how the distributions E_j , for $j > 1$, evolve. We find:

$$\frac{dE_j}{dt} = (1 - \theta_{n_f j}) \left[2n_f P_{qg} \otimes g + n_f P_{qq} \otimes \left(\frac{\Sigma}{6} + \sum_{i=n_f+1}^6 \frac{E_i}{i(i-1)} \right) \right] + \theta_{n_f j} P^+ \otimes E_j, \quad (141)$$

with:

$$\theta_{n_f j} = \begin{cases} 1 & n_f \geq j \\ 0 & n_f < j \end{cases}. \quad (142)$$

⁵We stick to the assumption that intrinsic contributions are symmetric upon charge conjugation. This allow us to forget about the valence sector that will always evolve multiplicatively.

Therefore, as expected, for a given n_f (and thus a given energy) the distribution E_j evolves multiplicatively through P^+ if $j \leq n_f$ and evolves exactly like the singlet if $j > n_f$. As an example, for $n_f = 4$, that is to say for energies between the charm and the bottom thresholds, $E_2 = T_3$, $E_3 = T_8$, and $E_4 = T_{15}$ evolve multiplicatively, while $E_5 = T_{24}$ and $E_6 = T_{35}$ evolve like the singlet.

Finally, the gluon distribution evolves as:

$$\frac{dg}{dt} = P_{gg} \otimes g + P_{gq} \otimes \sum_{i=1}^{n_f} q_i^+, \quad (143)$$

that translates into:

$$\frac{dg}{dt} = P_{gg} \otimes g + n_f P_{gq} \otimes \left(\frac{\Sigma}{6} + \sum_{j=n_f+1}^6 \frac{E_j}{j(j-1)} \right). \quad (144)$$

Eqs. (139), (141), and (144) fully define the splitting kernel matrix in the general case of possible presence of intrinsic heavy-quark contributions. Noticeably, these equations correctly reduce to the more familiar ones if these contributions are set to zero.

Therefore, the evolution of the vector of distributions E with n_f active flavours and possible intrinsic heavy-quark contributions takes the matricial form:

$$\frac{dE}{dt} = \mathbb{P}^{(n_f)} \otimes E, \quad (145)$$

where the entries of the splitting-function matrix $\mathbb{P}^{(n_f)}$ can be defined algorithmically as:

$$\mathbb{P}_{ij}^{(n_f)} = \begin{cases} \begin{array}{lll} P_{gg} & i=0 & j=0 \\ \frac{n_f}{6} P_{gq} & i=0 & j=1 \\ 0 & i=0 & 2 \leq j \leq n_f \\ \frac{6}{j(j-1)} \frac{n_f}{6} P_{gq} & i=0 & n_f+1 \leq j \leq 6 \end{array} \\ \begin{array}{lll} 2n_f P_{qg} & i=1 & j=0 \\ \frac{n_f}{6} P_{qq} & i=1 & j=1 \\ 0 & i=1 & 2 \leq j \leq n_f \\ \frac{6}{j(j-1)} \frac{n_f}{6} P_{qq} & i=1 & n_f+1 \leq j \leq 6 \end{array} \\ \begin{array}{lll} \delta_{ij} P^+ & 2 \leq i \leq n_f & 0 \leq j \leq 6 \end{array} \\ \begin{array}{lll} 2n_f P_{qg} & n_f+1 \leq i \leq 6 & j=0 \\ \frac{n_f}{6} P_{qq} & n_f+1 \leq i \leq 6 & j=1 \\ 0 & n_f+1 \leq i \leq 6 & 2 \leq j \leq n_f \\ \frac{6}{j(j-1)} \frac{n_f}{6} P_{qq} & n_f+1 \leq i \leq 6 & n_f+1 \leq j \leq 6 \end{array} \end{cases} \quad (146)$$

2.1 Evolution operator

We assume that the set of distributions E evolves through the operator Γ as:

$$E(t) = \Gamma(t, t_0) \otimes E(t_0). \quad (147)$$

In order to achieve an efficient computation of the evolution operator, we need to identify the general structure of Γ . Given the possible presence of heavy-quark thresholds in the evolution interval $[t_0, t]$, the operator Γ is in fact a product of operators. More specifically, it can be written as:

$$\Gamma(t, t_0) = \Gamma^{(N)}(t, t_N) \otimes \prod_{n_f=N-1}^0 \mathcal{M}^{(n_f)} \otimes \Gamma^{(n_f)}(t_{n_f+1} - \epsilon, t_{n_f}) \quad (148)$$

with $\mathcal{M}^{(n_f)} = T [\mathbb{I} + \mathbb{K}^{(n_f)}] T^{-1}$ given in Eq. (129) and $t > t_N > \dots > t_1 > t_0$, being $t_1, \dots, t_N$ the heavy-quark thresholds enclosed in the evolution interval. Using Eq. (145), each evolution operator $\mathbf{\Gamma}^{(n_f)}$ obeys the evolution equation:

$$\frac{d}{dt} \mathbf{\Gamma}^{(n_f)}(t, t_{n_f}) = \mathbb{P}^{(n_f)}(t) \otimes \mathbf{\Gamma}^{(n_f)}(t, t_{n_f}), \quad (149)$$

with boundary condition $\mathbf{\Gamma}^{(n_f)}(t_{n_f}, t_{n_f}) = \mathbb{I}$. The solution of the equation above can then be written as:

$$\mathbf{\Gamma}^{(n_f)}(t, t_{n_f}) = \mathcal{P} \exp \left[\int_{t_{n_f}}^t dt' \mathbb{P}^{(n_f)}(t') \right], \quad (150)$$

where $\mathcal{P}$ symbolises the path ordering. Given the structure of the splitting-function matrix in Eq. (146), one can infer the structure of $\mathbf{\Gamma}^{(n_f)}$. It turns out that the structure of $\mathbf{\Gamma}^{(n_f)}$ is almost the same as that of $\mathbb{P}^{(n_f)}$. Specifically:

$$\mathbf{\Gamma}_{ij}^{(n_f)} = \mathbb{I}_{ij} + \begin{cases} \begin{array}{lll} \Gamma_{gg} & i = 0 & j = 0 \\ \Gamma_{gq} & i = 0 & j = 1 \\ 0 & i = 0 & 2 \leq j \leq n_f \\ \frac{6}{j(j-1)} \Gamma_{gq} & i = 0 & n_f + 1 \leq j \leq 6 \end{array} \\ \begin{array}{lll} \Gamma_{qg} & i = 1 & j = 0 \\ \Gamma_{qq} & i = 1 & j = 1 \\ 0 & i = 1 & 2 \leq j \leq n_f \\ \frac{6}{j(j-1)} \Gamma_{qq} & i = 1 & n_f + 1 \leq j \leq 6 \end{array} \\ \delta_{ij} \Gamma^+ & 2 \leq i \leq n_f & 0 \leq j \leq 6 \\ \begin{array}{lll} \Gamma_{qg} & n_f + 1 \leq i \leq 6 & j = 0 \\ \Gamma_{qq} & n_f + 1 \leq i \leq 6 & j = 1 \\ 0 & n_f + 1 \leq i \leq 6 & 2 \leq j \leq n_f \\ \frac{6}{j(j-1)} \Gamma_{qq} & n_f + 1 \leq i \leq 6 & n_f + 1 \leq j \leq 6 \end{array} \end{cases} \cdot \quad (151)$$

The next step is the computation of the product $\mathcal{M}^{(n_f)} \mathbf{\Gamma}^{(n_f)} = T [\mathbb{I} + \mathbb{K}^{(n_f)}] T^{-1} \mathbf{\Gamma}^{(n_f)}$ appearing in Eq. (148) that we conveniently split as:

$$\mathcal{M}^{(n_f)} \mathbf{\Gamma}^{(n_f)} = \mathbf{\Gamma}^{(n_f)} + T \mathbb{K}^{(n_f)} T^{-1} + \mathbb{S}^{(n_f)}, \quad (152)$$

where we have defined $\mathbb{S}^{(n_f)} = T \mathbb{K}^{(n_f)} T^{-1} [\mathbf{\Gamma}^{(n_f)} - \mathbb{I}]$ that is given by:

$$\mathbb{S}_{ij}^{(n_f)} = \begin{cases} \begin{array}{lll} M_1 \Gamma_{gg} + \frac{1}{1+n_f} (M_2 - M_3) \Gamma_{gq} & i = 0 & j = 0 \\ M_1 \Gamma_{gq} + \frac{1}{1+n_f} (M_2 - M_3) \Gamma_{qq} & i = 0 & j = 1 \\ 0 & i = 0 & 2 \leq j \leq n_f \\ \frac{1}{j(j-1)} \left[M_1 \Gamma_{gq} + \frac{1}{1+n_f} (M_2 - M_3) \Gamma_{qq} \right] & i = 0 & n_f + 1 \leq j \leq 6 \end{array} \\ \begin{array}{lll} M_4 \Gamma_{gg} + \frac{1}{n_f+1} (M_5 - M_6) \Gamma_{qg} & i = 1 & j = 0 \\ M_4 \Gamma_{gq} + \frac{1}{n_f+1} (M_5 - M_6) \Gamma_{qq} & i = 1 & j = 1 \\ 0 & i = 1 & 2 \leq j \leq n_f \\ \frac{1}{j(j-1)} \left[M_4 \Gamma_{gq} + \frac{1}{n_f+1} (M_5 - M_6) \Gamma_{qq} \right] & i = 1 & n_f + 1 \leq j \leq 6 \end{array} \\ \delta_{ij} M_7 \Gamma^+ & 2 \leq i \leq n_f & 0 \leq j \leq 6 \\ \begin{array}{lll} -n_f \left(M_4 \Gamma_{gg} + \frac{1}{n_f+1} (M_5 - M_6) \Gamma_{qg} \right) + (n_f + 1) M_7 \Gamma_{qg} & i = n_f + 1 & j = 0 \\ -n_f \left(M_4 \Gamma_{gq} + \frac{1}{n_f+1} (M_5 - M_6) \Gamma_{qq} \right) + (n_f + 1) M_7 \Gamma_{qq} & i = n_f + 1 & j = 1 \\ 0 & i = n_f + 1 & 2 \leq j \leq n_f \\ \frac{6}{j(j-1)} \left[-n_f \left(M_4 \Gamma_{gq} + \frac{1}{n_f+1} (M_5 - M_6) \Gamma_{qq} \right) + (n_f + 1) M_7 \Gamma_{qq} \right] & i = n_f + 1 & n_f + 1 \leq j \leq 6 \end{array} \\ \begin{array}{lll} \frac{6}{j(j-1)} \left[M_4 \Gamma_{gq} + \frac{1}{n_f+1} (M_5 - M_6) \Gamma_{qq} \right] & n_f + 2 \leq i \leq 6 & n_f + 2 \leq j \leq 6 \end{array} \end{cases} \quad (153)$$

Eq. (152) along with Eq. (153) allow us to infer the general structure on the n_f -th factor in Eq. (148). It should be clear that the structure is driven by the term $T\mathbb{K}^{(n_f)}T^{-1}$ in Eq. (129).

We finally comment on FFs. So far, to the best of my knowledge, none of the $\mathcal{O}(\alpha_s^2)$ corrections to the FF matching conditions has been computed. Therefore, we must limit to $\mathbb{K}^{(1)(n_f)}$ that is simply given by the transpose of that for PDFs.

2.2 In the presence of N³LO corrections

If $\mathcal{O}(\alpha_s^3)$ corrections are included in the matching, matching conditions in Eq. (123) need to be generalised as follows:

$$\begin{aligned}
g^{(n_f+1)} &= [1 + K_{gg}]g^{(n_f)} + K_{gl} \sum_l l^{+(n_f)} + K_{gh}h^{+(n_f)}, \\
l^{+(n_f+1)} &= [1 + K_{ll}]l^{+(n_f)} + K_{ll'} \sum_l l^{+(n_f)} + K_{lg}g^{(n_f)}, \\
h^{+(n_f)} &= K_{hg}g^{(n_f)} + K_{hl} \sum_l l^{+(n_f)} + [1 + K_{hh}]h^{+(n_f)}, \\
H^{+(n_f+1)} &= H^{+(n_f)}, \\
l^{-(n_f+1)} &= [1 + K_{ll}^-]l^{-(n_f)},
\end{aligned} \tag{154}$$

where the new matching functions $K_{ll'}$ and K_{lg} , which start at $\mathcal{O}(\alpha_s^3)$, are introduced and where we neglected a possible K_{lh} that in principle exists but will not probably be computed in the next future. Also, we introduced the matching function K_{ll}^- which differs from K_{ll} starting from $\mathcal{O}(\alpha_s^3)$. With this modification, the matching algorithm in the evolution basis in Eq. (129) is more complicated. First we need to define two more combinations w.r.t. Eq. (128), *i.e.* M_8 and M_9 :

$$\begin{aligned}
M_1 &= K_{gg}, \\
M_2 &= K_{gh} + n_f K_{gl}, \\
M_3 &= K_{gh} - K_{gl}, \\
M_4 &= K_{hg} + n_f K_{lg}, \\
M_5 &= K_{hh} + n_f (K_{hl} + K_{ll} + n_f K_{ll'}), \\
M_6 &= K_{hh} - (K_{hl} + K_{ll} + n_f K_{ll'}), \\
M_7 &= K_{ll}, \\
M_8 &= K_{hg} - K_{lg}, \\
M_9 &= K_{hh} + n_f (K_{hl} + K_{ll} - K_{ll'}), \\
M_{10} &= K_{ll}^-,
\end{aligned} \tag{155}$$

which are such that up to NNLO $M_8 = M_4$, $M_9 = M_5$, and $M_{10} = M_7$. With these at hand, we find:

$$\left[T \mathbb{K}^{(n_f)} T^{-1} \right]_{ij} = \begin{cases} \begin{array}{lll} M_1 & i=0 & j=0 \\ \frac{1}{6} M_2 & i=0 & j=1 \\ 0 & i=0 & 2 \leq j \leq n_f \\ -\frac{1}{n_f+1} M_3 & i=0 & j=n_f+1 \\ \frac{1}{j(j-1)} M_2 & i=0 & n_f+2 \leq j \leq 6 \end{array} \\ \begin{array}{lll} M_4 & i=1 & j=0 \\ \frac{1}{6} M_5 & i=1 & j=1 \\ 0 & i=1 & 2 \leq j \leq n_f \\ -\frac{1}{n_f+1} M_6 & i=1 & j=n_f+1 \\ \frac{1}{j(j-1)} M_5 & i=1 & n_f+2 \leq j \leq 6 \end{array} \\ \delta_{ij} M_7 & 2 \leq i \leq n_f & 0 \leq j \leq 6 \\ \begin{array}{lll} -n_f M_8 & i=n_f+1 & j=0 \\ \frac{n_f}{6} (-M_9 + (n_f+1)M_7) & i=n_f+1 & j=1 \\ 0 & i=n_f+1 & 2 \leq j \leq n_f \\ \frac{1}{n_f+1} (n_f M_6 + (n_f+1)M_7 + M_5 - M_9) & i=n_f+1 & j=n_f+1 \\ \frac{n_f}{j(j-1)} (-M_9 + (n_f+1)M_7) & i=n_f+1 & n_f+2 \leq j \leq 6 \end{array} \\ \begin{array}{lll} M_4 & n_f+2 \leq i \leq 6 & j=0 \\ \frac{1}{6} M_5 & n_f+2 \leq i \leq 6 & j=1 \\ 0 & n_f+2 \leq i \leq 6 & 2 \leq j \leq n_f \\ -\frac{1}{n_f+1} M_6 & n_f+2 \leq i \leq 6 & j=n_f+1 \\ \frac{1}{j(j-1)} M_5 & n_f+2 \leq i \leq 6 & n_f+2 \leq j \leq 6 \end{array} \end{cases} \quad (156)$$

In the light of Eq. (154), valence and momentum sum rules imply the following identities for the matching functions:

$$\begin{aligned} \int_0^1 dy K_u^{-,[n]}(y) &= 0, \\ \int_0^1 dy y \left[K_{gg}^{[n]}(y) + n_f K_{lg}^{[n]}(y) + K_{hg}^{[n]}(y) \right] &= 0, \\ \int_0^1 dy y \left[K_{gl}^{[n]}(y) + K_{ll}^{[n]}(y) + n_f K_{ll'}^{[n]}(y) + K_{hl}^{[n]}(y) \right] &= 0, \\ \int_0^1 dy y \left[K_{gh}^{[n]}(y) + K_{hh}^{[n]}(y) \right] &= 0, \end{aligned} \quad (157)$$

valid at any give order n in the strong coupling α_s .

3 Evolution and matching in the physical basis

In view of the implementation of QED corrections and the ensuing complications due to the isospin symmetry breaking, we work out here how to implement PDF evolution and matching in pure QCD but in the physical basis. The purpose is to test the performance of this basis in **APFEL++** with the purpose of using this basis also in the presence of QED corrections. Rather than the actual physical basis $\{\bar{t}, \bar{b}, \dots, \bar{d}, g, d, \dots, b, t\}$, it is convenient to use the basis $\{f_i\} = \{t^-, b^-, \dots, d^-, g, d^+, \dots, b^+, t^+\}$, with $q^\pm = q \pm \bar{q}$, in that it guarantees a partial diagonalisation of the splitting-function matrix. In

this basis, we find that the evolution equations read:

$$\begin{aligned}
\frac{dq_k^-}{d \ln \mu^2} &= P^{\text{NV}} q_k^- + P^{\text{PV}} \sum_{i=1, i \neq k}^{n_f} q_i^-, \\
\frac{dg}{d \ln \mu^2} &= P_{gg} g + P_{gq} \sum_{i=1}^{n_f} q_i^+, \\
\frac{dq_k^+}{d \ln \mu^2} &= 2P_{qg} g + P^{\text{NS}} q_k^+ + P^{\text{PS}} \sum_{i=1, i \neq k}^{n_f} q_i^+,
\end{aligned} \tag{158}$$

where we have defined:

$$\begin{aligned}
P^{\text{PV}} &= \frac{1}{n_f} (P^V - P^-), & P^{\text{PS}} &= \frac{1}{n_f} (P_{qq} - P^+), \\
P^{\text{NV}} &= P^- + P^{\text{PV}}, & P^{\text{NS}} &= P^+ + P^{\text{PS}}.
\end{aligned} \tag{159}$$

and with $k = 1, \dots, n_f$ while:

$$\frac{dq_k^-}{d \ln \mu^2} = \frac{dq_k^+}{d \ln \mu^2} = 0, \quad \text{for } k = n_f + 1, \dots, 6. \tag{160}$$

Writing the full set of evolution equations as:

$$\frac{df_i}{d \ln \mu^2} = \mathbb{P}_{ij}^{(n_f)} f_j, \tag{161}$$

with $i, j = -6, \dots, 6$, one has that:

$$\mathbb{P}_{ij}^{(n_f)} = \begin{cases} P^{\text{PV}} & -n_f \leq i \leq -1 & -n_f \leq j \leq -1 \parallel j \neq i \\ P^{\text{NV}} & -n_f \leq i \leq -1 & j = i \\ P_{gg} & i = 0 & j = 0 \\ P_{gq} & i = 0 & 1 \leq j \leq n_f \\ 2P_{qg} & 1 \leq i \leq n_f & j = 0 \\ P^{\text{NS}} & 1 \leq i \leq n_f & j = i \\ P^{\text{PS}} & 1 \leq i \leq n_f & 1 \leq j \leq n_f \parallel j \neq i \\ 0 & \text{otherwise} \end{cases}. \tag{162}$$

As far as matching conditions are concerned, using Eq. (154), one immediately finds:

$$\mathbb{K}_{ij}^{(n_f)} = \begin{cases} K_{gg} & i = 0 & j = 0 \\ K_{gl} & i = 0 & 1 \leq j \leq n_f \\ K_{gh} & i = 0 & j = n_f + 1 \\ K_{lg} & 1 \leq i \leq n_f & j = 0 \\ K_{ll} & 1 \leq i \leq n_f & j = i \\ K_{ll'} & 1 \leq i \leq n_f & 1 \leq j \leq n_f \\ K_{hg} & i = n_f + 1 & j = 0 \\ K_{hl} & i = n_f + 1 & 1 \leq j \leq n_f \\ K_{hh} & i = n_f + 1 & j = n_f + 1 \\ 0 & \text{elsewhere} \end{cases}. \tag{163}$$

It is useful to establish a connection with the M 's in Eq. (155). Specifically we find:

$$\begin{aligned}
K_{gg} &= M_1, \\
K_{gl} &= \frac{M_2 - M_3}{n_f + 1}, \\
K_{gh} &= \frac{M_2 + n_f M_3}{n_f + 1}, \\
K_{lg} &= \frac{M_4 - M_8}{n_f + 1}, \\
K_{ll} &= M_7, \\
K_{ll'} &= \frac{M_5 - M_9}{n_f(n_f + 1)}, \\
K_{hg} &= \frac{M_4 + n_f M_8}{n_f + 1}, \\
K_{hl} &= \frac{M_9 - M_6}{n_f + 1} - M_7, \\
K_{hh} &= \frac{M_5 + n_f M_6}{n_f + 1}, \\
K_{ll}^- &= M_{10}.
\end{aligned} \tag{164}$$

4 Evolution and matching with QED corrections

In this section, we address the question of deriving the PDF evolution operator in the presence of QED corrections including both photon and charged-lepton PDFs. The fact that the photon coupling to quarks is proportional to the electric charge, while the QCD coupling is proportional to the strong coupling, complicates the description. The general form of the QCD+QED evolution equations is:

$$\frac{\partial}{\partial t} \begin{pmatrix} L_\alpha \\ U_j \\ D_i \\ \Gamma_x \\ \bar{D}_k \\ \bar{U}_h \\ \bar{L}_\beta \end{pmatrix} = \begin{pmatrix} \mathcal{P}_{L_\alpha L_\gamma} & \mathcal{P}_{L_\alpha U_e} & \mathcal{P}_{L_\alpha D_l} & \mathcal{P}_{L_\alpha \Gamma_y} & \mathcal{P}_{L_\alpha \bar{D}_m} & \mathcal{P}_{L_\alpha \bar{U}_n} & \mathcal{P}_{L_\alpha \bar{L}_\delta} \\ \mathcal{P}_{U_j L_\gamma} & \mathcal{P}_{U_j U_e} & \mathcal{P}_{U_j D_l} & \mathcal{P}_{U_j \Gamma_y} & \mathcal{P}_{U_j \bar{D}_m} & \mathcal{P}_{U_j \bar{U}_n} & \mathcal{P}_{U_j \bar{L}_\delta} \\ \mathcal{P}_{D_i L_\gamma} & \mathcal{P}_{D_i U_e} & \mathcal{P}_{D_i D_l} & \mathcal{P}_{D_i \Gamma_y} & \mathcal{P}_{D_i \bar{D}_m} & \mathcal{P}_{D_i \bar{U}_n} & \mathcal{P}_{D_i \bar{L}_\delta} \\ \mathcal{P}_{\Gamma_x L_\gamma} & \mathcal{P}_{\Gamma_x U_e} & \mathcal{P}_{\Gamma_x D_l} & \mathcal{P}_{\Gamma_x \Gamma_y} & \mathcal{P}_{\Gamma_x \bar{D}_m} & \mathcal{P}_{\Gamma_x \bar{U}_n} & \mathcal{P}_{\Gamma_x \bar{L}_\delta} \\ \mathcal{P}_{\bar{D}_k L_\gamma} & \mathcal{P}_{\bar{D}_k U_e} & \mathcal{P}_{\bar{D}_k D_l} & \mathcal{P}_{\bar{D}_k \Gamma_y} & \mathcal{P}_{\bar{D}_k \bar{D}_m} & \mathcal{P}_{\bar{D}_k \bar{U}_n} & \mathcal{P}_{\bar{D}_k \bar{L}_\delta} \\ \mathcal{P}_{\bar{U}_h L_\gamma} & \mathcal{P}_{\bar{U}_h U_e} & \mathcal{P}_{\bar{U}_h D_l} & \mathcal{P}_{\bar{U}_h \Gamma_y} & \mathcal{P}_{\bar{U}_h \bar{D}_m} & \mathcal{P}_{\bar{U}_h \bar{U}_n} & \mathcal{P}_{\bar{U}_h \bar{L}_\delta} \\ \mathcal{P}_{\bar{L}_\beta L_\gamma} & \mathcal{P}_{\bar{L}_\beta U_e} & \mathcal{P}_{\bar{L}_\beta D_l} & \mathcal{P}_{\bar{L}_\beta \Gamma_y} & \mathcal{P}_{\bar{L}_\beta \bar{D}_m} & \mathcal{P}_{\bar{L}_\beta \bar{U}_n} & \mathcal{P}_{\bar{L}_\beta \bar{L}_\delta} \end{pmatrix} \begin{pmatrix} L_\gamma \\ U_e \\ D_l \\ \Gamma_y \\ \bar{D}_m \\ \bar{U}_n \\ \bar{L}_\delta \end{pmatrix}, \tag{165}$$

with $t = \ln \mu^2$ and where $D_i \in \{D_d, D_s, D_b\}$, $U_j \in \{U_u, U_c, U_t\}$, $L_\alpha \in \{L_{e-}, L_{\mu-}, L_{\tau-}\}$ (and respective anti-flavours such that $\bar{X}_i = X_i^-$), and $\Gamma_x \in \{g, \gamma\}$. A sum over repeated indices is also understood. Importantly, each distribution X_i is in a given colour state, such that N_X copies of it exist, where N_X is the number of possible colour states. Clearly, we have that $N_D = N_U = N_c = 3$ and $N_L = 1$. A decomposition of the splitting functions in terms of valence and singlet contributions is as follows:

$$\begin{aligned}
\mathcal{P}_{X_i Y_j} &= \mathcal{P}_{\bar{X}_i \bar{Y}_j} = \delta_{ij} \mathcal{P}_{XX}^V + \mathcal{P}_{XY}^S, \\
\mathcal{P}_{\bar{X}_i Y_j} &= \mathcal{P}_{X_i \bar{Y}_j} = \delta_{ij} \mathcal{P}_{XX}^V + \mathcal{P}_{XY}^S, \\
\mathcal{P}_{X_i \Gamma_x} &= \mathcal{P}_{\bar{X}_i \Gamma_x} = \mathcal{P}_{X\Gamma_x}, \\
\mathcal{P}_{\Gamma_x X_i} &= \mathcal{P}_{\Gamma_x \bar{X}_i} = \mathcal{P}_{\Gamma_x X}.
\end{aligned} \tag{166}$$

where $X, Y \in \{D, U, L\}$. This decomposition implies that:

$$\begin{aligned} \frac{\partial}{\partial t} \begin{pmatrix} L_\alpha \\ U_j \\ D_i \\ \Gamma_x \\ \bar{D}_k \\ \bar{U}_h \\ \bar{L}_\beta \end{pmatrix} &= \begin{pmatrix} \mathcal{P}_{LL}^V & 0 & 0 & \mathcal{P}_{L\Gamma_y} & 0 & 0 & \mathcal{P}_{LL}^V \\ 0 & \mathcal{P}_{UU}^V & 0 & \mathcal{P}_{U\Gamma_y} & 0 & \mathcal{P}_{UU}^V & 0 \\ 0 & 0 & \mathcal{P}_{DD}^V & \mathcal{P}_{D\Gamma_y} & \mathcal{P}_{DD}^V & 0 & 0 \\ \mathcal{P}_{\Gamma_x L} & \mathcal{P}_{\Gamma_x U} & \mathcal{P}_{\Gamma_x D} & \mathcal{P}_{\Gamma_x \Gamma_y} & \mathcal{P}_{\Gamma_x D} & \mathcal{P}_{\Gamma_x U} & \mathcal{P}_{\Gamma_x L} \\ 0 & 0 & \mathcal{P}_{\bar{D}D}^V & \mathcal{P}_{D\Gamma_y} & \mathcal{P}_{\bar{D}D}^V & 0 & 0 \\ 0 & \mathcal{P}_{\bar{U}U}^V & 0 & \mathcal{P}_{U\Gamma_y} & 0 & \mathcal{P}_{\bar{U}U}^V & 0 \\ \mathcal{P}_{\bar{L}L}^V & 0 & 0 & \mathcal{P}_{L\Gamma_y} & 0 & 0 & \mathcal{P}_{\bar{L}L}^V \end{pmatrix} \begin{pmatrix} L_\alpha \\ U_j \\ D_i \\ \Gamma_y \\ \bar{D}_k \\ \bar{U}_h \\ \bar{L}_\beta \end{pmatrix} \\ &+ \begin{pmatrix} \mathcal{P}_{LL}^S & \mathcal{P}_{UL}^S & \mathcal{P}_{DL}^S & 0 & \mathcal{P}_{DL}^S & \mathcal{P}_{UL}^S & \mathcal{P}_{LL}^S \\ \mathcal{P}_{UL}^S & \mathcal{P}_{UU}^S & \mathcal{P}_{DU}^S & 0 & \mathcal{P}_{DU}^S & \mathcal{P}_{UU}^S & \mathcal{P}_{UL}^S \\ \mathcal{P}_{DL}^S & \mathcal{P}_{DU}^S & \mathcal{P}_{DD}^S & 0 & \mathcal{P}_{DD}^S & \mathcal{P}_{DU}^S & \mathcal{P}_{DL}^S \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \mathcal{P}_{DL}^S & \mathcal{P}_{DU}^S & \mathcal{P}_{DD}^S & 0 & \mathcal{P}_{DD}^S & \mathcal{P}_{DU}^S & \mathcal{P}_{DL}^S \\ \mathcal{P}_{UL}^S & \mathcal{P}_{UU}^S & \mathcal{P}_{DU}^S & 0 & \mathcal{P}_{DU}^S & \mathcal{P}_{UU}^S & \mathcal{P}_{UL}^S \\ \mathcal{P}_{LL}^S & \mathcal{P}_{UL}^S & \mathcal{P}_{DL}^S & 0 & \mathcal{P}_{DL}^S & \mathcal{P}_{UL}^S & \mathcal{P}_{LL}^S \end{pmatrix} \begin{pmatrix} \sum_\gamma L_\gamma \\ \sum_e U_e \\ \sum_l D_l \\ \Gamma_y \\ \sum_m \bar{D}_m \\ \sum_n \bar{U}_n \\ \sum_\delta \bar{L}_\delta \end{pmatrix}. \end{aligned} \quad (167)$$

Now, we define $X_i^\pm = X_i \pm \bar{X}_i$, which implies:

$$\begin{aligned} \frac{\partial}{\partial t} \begin{pmatrix} L_\alpha^+ \\ U_j^+ \\ D_i^+ \\ \Gamma_x \\ D_i^- \\ U_j^- \\ L_\alpha^- \end{pmatrix} &= \begin{pmatrix} \mathcal{P}_{LL}^+ & 0 & 0 & 2\mathcal{P}_{L\Gamma_y} & 0 & 0 & 0 \\ 0 & \mathcal{P}_{UU}^+ & 0 & 2\mathcal{P}_{U\Gamma_y} & 0 & 0 & 0 \\ 0 & 0 & \mathcal{P}_{DD}^+ & 2\mathcal{P}_{D\Gamma_y} & 0 & 0 & 0 \\ \mathcal{P}_{\Gamma_x L} & \mathcal{P}_{\Gamma_x U} & \mathcal{P}_{\Gamma_x D} & \mathcal{P}_{\Gamma_x \Gamma_y} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathcal{P}_{DD}^- & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathcal{P}_{UU}^- & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \mathcal{P}_{LL}^- \end{pmatrix} \begin{pmatrix} L_\alpha^+ \\ U_j^+ \\ D_i^+ \\ \Gamma_y \\ D_i^- \\ U_j^- \\ L_\alpha^- \end{pmatrix} \\ &+ \begin{pmatrix} \mathcal{P}_{LL}^{\text{PS}} & \mathcal{P}_{LU}^{\text{PS}} & \mathcal{P}_{LD}^{\text{PS}} & 0 & 0 & 0 & 0 \\ \mathcal{P}_{UL}^{\text{PS}} & \mathcal{P}_{UU}^{\text{PS}} & \mathcal{P}_{UD}^{\text{PS}} & 0 & 0 & 0 & 0 \\ \mathcal{P}_{DL}^{\text{PS}} & \mathcal{P}_{DU}^{\text{PS}} & \mathcal{P}_{DD}^{\text{PS}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathcal{P}_{DD}^{\text{PV}} & \mathcal{P}_{DU}^{\text{PV}} & \mathcal{P}_{DL}^{\text{PV}} \\ 0 & 0 & 0 & 0 & \mathcal{P}_{UD}^{\text{PV}} & \mathcal{P}_{UU}^{\text{PV}} & \mathcal{P}_{UL}^{\text{PV}} \\ 0 & 0 & 0 & 0 & \mathcal{P}_{LD}^{\text{PV}} & \mathcal{P}_{LU}^{\text{PV}} & \mathcal{P}_{LL}^{\text{PV}} \end{pmatrix} \begin{pmatrix} \sum_\gamma^{n_L} L_\gamma^+ \\ \sum_\gamma^{n_U} U_\gamma^+ \\ \sum_e^{n_U} U_e^+ \\ \sum_l^{n_D} D_l^+ \\ \Gamma_y \\ \sum_l^{n_D} D_l^- \\ \sum_\gamma^{n_U} U_\gamma^- \\ \sum_\gamma^{n_L} L_\gamma^- \end{pmatrix}, \end{aligned} \quad (168)$$

where we have defined:

$$\mathcal{P}_{XX}^\pm = \mathcal{P}_{XX}^V \pm \mathcal{P}_{\bar{X}\bar{X}}^V, \quad \mathcal{P}_{XY}^{\text{PS}} = \mathcal{P}_{XY}^S + \mathcal{P}_{\bar{X}\bar{Y}}^S, \quad \mathcal{P}_{XY}^{\text{PV}} = \mathcal{P}_{XY}^S - \mathcal{P}_{\bar{X}\bar{Y}}^S. \quad (169)$$

Also notice that the sums over flavours run up to n_D , n_U , and n_L , which are the number of active down-type flavours, up-type flavours, and leptons, respectively, and over all possible colour states (despite not explicitly indicated). We can now start scrutinising the splitting functions. First, their perturbative expansion reads:

$$\mathcal{P} = \underbrace{a_s \mathcal{P}^{[1,0]} + a \mathcal{P}^{[0,1]}}_{\text{LO}} + \underbrace{a_s^2 \mathcal{P}^{[2,0]} + a_s a \mathcal{P}^{[1,1]} + a_s^2 \mathcal{P}^{[0,2]}}_{\text{NLO}} + \underbrace{a_s^3 \mathcal{P}^{[3,0]} + a_s^4 \mathcal{P}^{[4,0]}}_{\text{NNLO} + \text{N}^3\text{LO (QCD only)}}, \quad (170)$$

with $a_s = \alpha_s/(4\pi)$ and $a = \alpha/(4\pi)$ and where we have grouped the perturbative terms according to the cumulative powers of a_s and a . Also, we are interested in QED corrections up to two loops, *i.e.* NLO. Higher orders are only accounted for in QCD and are treated as discussed in the previous section. In this approximation, the pure-valence block ($\mathcal{P}^{\text{PV}}$), which starts at NNLO, is purely given by QCD so that:

$$\mathcal{P}_{DL}^{\text{PV}} = \mathcal{P}_{LD}^{\text{PV}} = \mathcal{P}_{UL}^{\text{PV}} = \mathcal{P}_{LU}^{\text{PV}} = \mathcal{P}_{LL}^{\text{PV}} = 0, \quad (171)$$

and:

$$\mathcal{P}_{DD}^{\text{PV}} = \mathcal{P}_{DU}^{\text{PV}} = \mathcal{P}_{DD}^{\text{PV}} = \mathcal{P}^{\text{PV}}. \quad (172)$$

Moreover, at this order there is no coupling between leptons and gluon, such that:

$$\mathcal{P}_{Lg} = \mathcal{P}_{gL} = 0, \quad (173)$$

so that, we finally have:

$$\begin{aligned} \frac{\partial}{\partial t} \begin{pmatrix} L_\alpha^+ \\ U_j^+ \\ D_i^+ \\ g \\ \gamma \\ D_i^- \\ U_j^- \\ L_\alpha^- \end{pmatrix} &= \begin{pmatrix} \mathcal{P}_{LL}^+ & 0 & 0 & 0 & 2\mathcal{P}_{L\gamma} & 0 & 0 & 0 \\ 0 & \mathcal{P}_{UU}^+ & 0 & 2\mathcal{P}_{Ug} & 2\mathcal{P}_{U\gamma} & 0 & 0 & 0 \\ 0 & 0 & \mathcal{P}_{DD}^+ & 2\mathcal{P}_{Dg} & 2\mathcal{P}_{D\gamma} & 0 & 0 & 0 \\ 0 & \mathcal{P}_{gU} & \mathcal{P}_{gD} & \mathcal{P}_{gg} & \mathcal{P}_{g\gamma} & 0 & 0 & 0 \\ \mathcal{P}_{\gamma L} & \mathcal{P}_{\gamma U} & \mathcal{P}_{\gamma D} & \mathcal{P}_{\gamma g} & \mathcal{P}_{\gamma\gamma} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathcal{P}_{DD}^- & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \mathcal{P}_{UU}^- & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathcal{P}_{LL}^- \end{pmatrix} \begin{pmatrix} L_\alpha^+ \\ U_j^+ \\ D_i^+ \\ g \\ \gamma \\ D_i^- \\ U_j^- \\ L_\alpha^- \end{pmatrix} \\ &+ \begin{pmatrix} \mathcal{P}_{LL}^{\text{PS}} & \mathcal{P}_{LU}^{\text{PS}} & \mathcal{P}_{LD}^{\text{PS}} & 0 & 0 & 0 & 0 & 0 \\ \mathcal{P}_{UL}^{\text{PS}} & \mathcal{P}_{UU}^{\text{PS}} & \mathcal{P}_{UD}^{\text{PS}} & 0 & 0 & 0 & 0 & 0 \\ \mathcal{P}_{DL}^{\text{PS}} & \mathcal{P}_{DU}^{\text{PS}} & \mathcal{P}_{DD}^{\text{PS}} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & P^{\text{PV}} & P^{\text{PV}} & 0 \\ 0 & 0 & 0 & 0 & 0 & P^{\text{PV}} & P^{\text{PV}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \sum_{\gamma}^{n_L} L_\gamma^+ \\ \sum_{e}^{n_U} U_e^+ \\ \sum_l^{n_D} D_l^+ \\ g \\ \gamma \\ \sum_l^{n_D} D_l^- \\ \sum_{e}^{n_U} U_e^- \\ \sum_{\gamma}^{n_L} L_\gamma^- \end{pmatrix}, \end{aligned} \quad (174)$$

It remains to assign to each non-zero entry of the splitting-function matrices the correct value, which we will do below. In view of this analysis, the basis that provides a partial diagonalisation of the evolution equation along with a relative simple implementation is:

$$\{f_i\} = \{L_\tau^+, L_\mu^+, L_e^+, U_t^+, U_c^+, U_u^+, D_b^+, D_s^+, D_d^+, g, \gamma, D_d^-, D_s^-, D_b^-, U_u^-, U_c^-, U_t^-, L_e^-, L_\mu^-, L_\tau^-\}. \quad (175)$$

In principle, one could establish a basis that maximally diagonalises the system (as for instance done in `QCD_QED_common_basis.pdf`). However, this basis is particularly cumbersome to handle.

Let us now turn to threshold matching. In order to implement it, one needs to be able establish what are the light flavours, what is the heavy flavour, and what are the super-heavy flavours. To this purpose, it is necessary to order distributions according to the thresholds of the corresponding partons. Unfortunately, at variance with the pure QCD case, this ordering is not reflected in the basis in Eq. (175). Given the ordered vector of thresholds $\{\mu_i\}$, one can order the distributions accordingly:

$$\{\tilde{f}_i\} = \{X_i^+, \Gamma_x, X_i^-\}, \quad (176)$$

where a possible realistic ordering could be:

$$\{\tilde{f}_i\} = \{U_t^+, D_b^+, L_\tau^+, U_c^+, L_\mu^+, D_s^+, D_d^+, U_u^+, L_e^+, g, \gamma, L_e^-, U_u^-, D_d^-, D_s^-, L_\mu^-, U_c^-, L_\tau^-, D_b^-, U_t^-\}, \quad (177)$$

which can however be determined dynamically according to the values of thresholds. With this new basis, the matching at the n_T -th mass threshold can be implemented sequentially, as in the case of pure QCD. Using l ($1 \leq l \leq n_T$), h ($= n_T + 1$), and H ($n_T + 2 \leq H \leq 9$) to indicate the light, heavy, and super-heavy flavours, respectively, the matching is implemented through the system:⁶

$$\tilde{f}_i^{(n_T+1)} = [\delta_{ij} + \mathcal{K}_{ij}(\kappa_h)] \tilde{f}_j^{(n_T)}. \quad (178)$$

with $\kappa_h = \mu_h/m_h$, being m_h the mass (not the threshold) of the heavy flavour h , and where:

$$\tilde{f}_i^{(n_T)} = \{H^{+(n_T)}, h^{+(n_T)}, l^{+(n_T)}, \Gamma_x^{(n_T)}, l^{-(n_T)}, h^{-(n_T)}, H^{-(n_T)}\}, \quad (179)$$

⁶It should be noted that, once the ordering in Eq. (176) has been established on the basis of thresholds, the value of n_T uniquely defines the values of n_D , n_U and n_L (which are in turn such that $n_D + n_U + n_L = n_T$).

and similarly for $\tilde{f}_i^{(n_T+1)}$. Given the transformation matrix T , such that $f_i = T_{ij}\tilde{f}_j$,⁷ one also has:

$$f_i^{(n_T+1)} = \left[\delta_{ij} + T_{ia}\mathcal{K}_{ab}(\kappa_h)T_{bj}^{-1} \right] f_j^{(n_T)}. \quad (180)$$

One explicitly has:

$$\begin{aligned} H^{+(n_T+1)} &= H^{+(n_T)}, \\ h^{+(n_T+1)} &= \sum_y \mathcal{K}_{hy}\Gamma_y^{(n_T)} + \sum_{l'} \mathcal{K}_{hl'}l'^{+(n_T)} + [1 + \mathcal{K}_{hh}]h^{+(n_T)}, \\ l^{+(n_T+1)} &= \sum_y \mathcal{K}_{ly}\Gamma_y^{(n_T)} + [1 + \mathcal{K}_{\text{NS},l}]l^{+(n_T)} + \sum_{l'} \mathcal{K}_{ll'}l'^{+(n_T)}, \\ \Gamma_x^{(n_T+1)} &= \sum_y [\delta_{xy} + \mathcal{K}_{xy}]\Gamma_y^{(n_T)} + \sum_{l'} \mathcal{K}_{xl'}l'^{+(n_T)} + \mathcal{K}_{xh}h^{+(n_T)}, \\ l^{-(n_T+1)} &= [1 + \mathcal{K}_{\text{NS},l}^-]l^{-(n_T)}, \\ h^{-(n_T+1)} &= h^{-(n_T)}, \\ H^{-(n_T+1)} &= H^{-(n_T)}, \end{aligned} \quad (181)$$

where, for brevity, we omitted the dependence of κ_h of the matching $\mathcal{K}$. Also, we notice that $\mathcal{K}_{\text{NS},l}^-$ differs from $\mathcal{K}_{\text{NS},l}$ starting from $\mathcal{O}(\alpha_s^3)$.

As in the case of splitting functions, we will include QED corrections up to NLO accuracy, leaving QCD only at higher orders. The perturbative expansion of the matching matrix will thus be:

$$\mathcal{K}_{ij} = \underbrace{a_s \mathcal{K}_{ij}^{[1,0]} + a \mathcal{K}_{ij}^{[0,1]}}_{\text{NLO}} + \underbrace{a_s^2 \mathcal{K}_{ij}^{[2,0]} + a_s^3 \mathcal{K}_{ij}^{[3,0]}}_{\text{NNLO} + \text{N}^3\text{LO (QCD only)}}. \quad (182)$$

Under this assumption, the structure of matching greatly simplifies. Indeed, up to NLO, which is the order up which QED corrections are included, the only matching functions to be different from zero are $\mathcal{K}_{xy}(\propto \delta_{xy})$, $\mathcal{K}_{xh}$, $\mathcal{K}_{hy}$, and $\mathcal{K}_{hh}$, all others start at NNLO or beyond. Since corrections beyond NLO are QCD only, they do not carry any flavour dependence and any function that involves leptons

⁷Note that T will be a sparse matrix with only one entry equal to one for each row and for each column but not along the diagonal.

and/or photon is identically zero. This leads to:

$$\begin{aligned}
H^{+(n_T+1)} &= H^{+(n_T)}, \\
h^{+(n_T+1)} &= \mathcal{K}_{hg}g^{(n_T)} + \mathcal{K}_{h\gamma}\gamma^{(n_T)} + \mathcal{K}_{hl'} \sum_{l'} l'^{+(n_T)} + [1 + \mathcal{K}_{hh}]h^{+(n_T)}, \\
l^{+(n_T+1)} &= \mathcal{K}_{lg}g^{(n_T)} + [1 + \mathcal{K}_{\text{NS},l}]l^{+(n_T)} + \mathcal{K}_{ll'} \sum_{l'} l'^{+(n_T)}, \\
g^{(n_T+1)} &= [1 + \mathcal{K}_{gg}]g^{(n_T)} + \mathcal{K}_{gl'} \sum_{l'} l'^{+(n_T)} + \mathcal{K}_{gh}h^{+(n_T)}, \\
\gamma^{(n_T+1)} &= [1 + \mathcal{K}_{\gamma\gamma}]\gamma^{(n_T)} + \mathcal{K}_{\gamma h}h^{+(n_T)}, \\
l^{-(n_T+1)} &= [1 + \mathcal{K}_{\text{NS},l}^-]l^{-(n_T)}, \\
h^{-(n_T+1)} &= h^{-(n_T)}, \\
H^{-(n_T+1)} &= H^{-(n_T)}.
\end{aligned} \tag{183}$$

If h is a lepton, such that $h = L_\alpha$, we have that all distributions of the basis f_i match on themselves ($f_i^{(n_T+1)} = f_i^{(n_T)}$), except:

$$\begin{aligned}
L_\alpha^{+(n_T+1)} &= [1 + \mathcal{K}_{LL}]L_\alpha^{+(n_T)} + \mathcal{K}_{L\gamma}\gamma^{(n_T)}, \\
\gamma^{(n_T+1)} &= \mathcal{K}_{\gamma L}L_\alpha^{+(n_T)} + [1 + \mathcal{K}_{\gamma\gamma}^L]\gamma^{(n_T)},
\end{aligned} \tag{184}$$

where all functions $\mathcal{K}$ implicitly depend on the variable κ_{L_α} . If instead $h = X_i = D_i, U_i$, we would still have $f_i^{(n_T+1)} = f_i^{(n_T)}$ for all but:

$$\begin{aligned}
X_i^{+(n_T+1)} &= \mathcal{K}_{Qg}g^{(n_T)} + \mathcal{K}_{X\gamma}\gamma^{(n_T)} + \mathcal{K}_{Qq'} \left(\sum_{j=1}^{n_D} D_j^{+(n_T)} + \sum_{j=1}^{n_U} U_j^{+(n_T)} \right) + [1 + \mathcal{K}_{XX}]X_i^{+(n_T)}, \\
q^{+(n_T+1)} &= \mathcal{K}_{qg}g^{(n_T)} + [1 + \mathcal{K}_{\text{NS},q}]q^{+(n_T)} + \mathcal{K}_{qq'} \left(\sum_{j=1}^{n_D} D_j^{+(n_T)} + \sum_{j=1}^{n_U} U_j^{+(n_T)} \right), \\
g^{(n_T+1)} &= [1 + \mathcal{K}_{gg}]g^{(n_T)} + \mathcal{K}_{gq} \left(\sum_{j=1}^{n_D} D_j^{+(n_T)} + \sum_{j=1}^{n_U} U_j^{+(n_T)} \right) + \mathcal{K}_{gQ}X_i^{+(n_T)}, \\
\gamma^{(n_T+1)} &= [1 + \mathcal{K}_{\gamma\gamma}^X]\gamma^{(n_T)} + \mathcal{K}_{\gamma X}X_i^{+(n_T)}, \\
q^{-(n_T+1)} &= [1 + \mathcal{K}_{\text{NS},q}^-]q^{-(n_T)},
\end{aligned} \tag{185}$$

where all functions $\mathcal{K}$ implicitly depend on the variable κ_{X_i} and where we used the character Q (q) to mean a generic heavy (light) quark, no matter whether of up or down type.

In fact, Eqs. (184) and (185) can be put together by allowing the index X in Eq. (185) to run over all possible values ($X = D, U, L$) and by setting:

$$\mathcal{K}_{Qg} = \mathcal{K}_{Qq'} = \mathcal{K}_{qg} = \mathcal{K}_{\text{NS},q} = \mathcal{K}_{qq'} = \mathcal{K}_{gg} = \mathcal{K}_{gq} = \mathcal{K}_{gQ} = 0, \tag{186}$$

when $h = L_\alpha$.

We are now in a position to give the specific form of splitting and matching functions. Let us start with splitting functions. We have 30 splitting functions to define, they are:

$$\begin{aligned}
& \mathcal{P}_{DD}^+, \mathcal{P}_{UU}^+, \mathcal{P}_{LL}^+, \\
& \mathcal{P}_{DD}^-, \mathcal{P}_{UU}^-, \mathcal{P}_{LL}^-, \\
& \mathcal{P}_{LL}^{\text{PS}}, \mathcal{P}_{LU}^{\text{PS}}, \mathcal{P}_{LD}^{\text{PS}}, \mathcal{P}_{UL}^{\text{PS}}, \mathcal{P}_{UU}^{\text{PS}}, \mathcal{P}_{UD}^{\text{PS}}, \mathcal{P}_{DL}^{\text{PS}}, \mathcal{P}_{DU}^{\text{PS}}, \mathcal{P}_{DD}^{\text{PS}}, \\
& P^{\text{PV}}, \\
& \mathcal{P}_{Dg}, \mathcal{P}_{Ug}, \mathcal{P}_{D\gamma}, \mathcal{P}_{U\gamma}, \mathcal{P}_{L\gamma}, \\
& \mathcal{P}_{gD}, \mathcal{P}_{gU}, \mathcal{P}_{\gamma D}, \mathcal{P}_{\gamma U}, \mathcal{P}_{\gamma L}, \\
& \mathcal{P}_{gg}, \mathcal{P}_{g\gamma}, \mathcal{P}_{\gamma g}, \mathcal{P}_{\gamma\gamma}.
\end{aligned} \tag{187}$$

At $\mathcal{O}(\alpha_s)$, we have that:

$$\begin{aligned}
& \mathcal{P}_{LL}^{+, [1,0]} = \mathcal{P}_{LL}^{-, [1,0]} = \mathcal{P}_{LL}^{\text{PS}, [1,0]} = \mathcal{P}_{LU}^{\text{PS}, [1,0]} = \mathcal{P}_{LD}^{\text{PS}, [1,0]} = \mathcal{P}_{UL}^{\text{PS}, [1,0]} = \mathcal{P}_{UU}^{\text{PS}, [1,0]} = \mathcal{P}_{UD}^{\text{PS}, [1,0]} = \mathcal{P}_{DL}^{\text{PS}, [1,0]} = \mathcal{P}_{DD}^{\text{PS}, [1,0]} = \\
& \mathcal{P}_{DU}^{\text{PS}, [1,0]} = P^{\text{PV}, [1,0]} = \mathcal{P}_{D\gamma}^{[1,0]} = \mathcal{P}_{U\gamma}^{[1,0]} = \mathcal{P}_{L\gamma}^{[1,0]} = \mathcal{P}_{\gamma D}^{[1,0]} = \mathcal{P}_{\gamma U}^{[1,0]} = \mathcal{P}_{\gamma L}^{[1,0]} = \mathcal{P}_{g\gamma}^{[1,0]} = \mathcal{P}_{\gamma g}^{[1,0]} = \mathcal{P}_{\gamma\gamma}^{[1,0]} = 0,
\end{aligned} \tag{188}$$

and:

$$\begin{aligned}
& \mathcal{P}_{DD}^{+, [1,0]} = \mathcal{P}_{UU}^{+, [1,0]} = \mathcal{P}_{DD}^{-, [1,0]} = \mathcal{P}_{UU}^{-, [1,0]} = 2C_F \left(\frac{1+x^2}{1-x} \right)_+, \\
& \mathcal{P}_{Dg}^{[1,0]} = \mathcal{P}_{Ug}^{[1,0]} = 2T_R [x^2 + (1-x)^2], \\
& \mathcal{P}_{gD}^{[1,0]} = \mathcal{P}_{gU}^{[1,0]} = 2C_F \frac{1+(1-x)^2}{x}, \\
& \mathcal{P}_{gg}^{[1,0]} = 4C_A \left[\frac{x}{(1-x)_+} + \frac{1-x}{x} + x(1-x) \right] + \beta_0 \delta(1-x).
\end{aligned} \tag{189}$$

At $\mathcal{O}(\alpha)$, we have an analogous picture:

$$\begin{aligned}
& \mathcal{P}_{LL}^{\text{PS}, [0,1]} = \mathcal{P}_{LU}^{\text{PS}, [0,1]} = \mathcal{P}_{LD}^{\text{PS}, [0,1]} = \mathcal{P}_{UL}^{\text{PS}, [0,1]} = \mathcal{P}_{UU}^{\text{PS}, [0,1]} = \mathcal{P}_{UD}^{\text{PS}, [0,1]} = \mathcal{P}_{DL}^{\text{PS}, [0,1]} = \mathcal{P}_{DU}^{\text{PS}, [0,1]} = \mathcal{P}_{DD}^{\text{PS}, [0,1]} = \\
& P^{\text{PV}, [0,1]} = \mathcal{P}_{Dg}^{[0,1]} = \mathcal{P}_{Ug}^{[0,1]} = \mathcal{P}_{gD}^{[0,1]} = \mathcal{P}_{gU}^{[0,1]} = \mathcal{P}_{\gamma D}^{[0,1]} = \mathcal{P}_{\gamma U}^{[0,1]} = \mathcal{P}_{\gamma L}^{[0,1]} = 0,
\end{aligned} \tag{190}$$

and:

$$\begin{aligned}
& \mathcal{P}_{XX}^{+, [0,1]} = \mathcal{P}_{XX}^{-, [0,1]} = 2e_X^2 \left(\frac{1+x^2}{1-x} \right)_+, \\
& \mathcal{P}_{X\gamma}^{[0,1]} = 2e_X^2 N_X [x^2 + (1-x)^2], \\
& \mathcal{P}_{\gamma X}^{[0,1]} = 2e_X^2 \frac{1+(1-x)^2}{x}, \\
& \mathcal{P}_{\gamma\gamma}^{[0,1]} = -\frac{4}{3} e_\Sigma^2 \delta(1-x),
\end{aligned} \tag{191}$$

for $X = D, U, L$, with the charges being $e_D^2 = 1/9$, $e_U^2 = 4/9$, $e_L^2 = 1$, and where we have defined:⁸

$$e_\Sigma^\alpha = \sum_{X=D,U,L} n_X N_X e_X^\alpha. \tag{192}$$

Now let us move to two loops considering first $\mathcal{O}(\alpha_s^2)$. We will refrain from writing the explicit

⁸Note that in Eq. (191) we have that $\alpha = 2$, but below we will also need $\alpha = 4$.

expressions for the splitting functions and refer to the usual QCD ones. At this order we have:

$$\begin{aligned} \mathcal{P}_{LL}^{+, [2,0]} &= \mathcal{P}_{LL}^{-, [2,0]} = \mathcal{P}_{LL}^{\text{PS}, [2,0]} = \mathcal{P}_{LU}^{\text{PS}, [2,0]} = \mathcal{P}_{LD}^{\text{PS}, [2,0]} = \mathcal{P}_{UL}^{\text{PS}, [2,0]} = \mathcal{P}_{DL}^{\text{PS}, [2,0]} = P^{\text{PV}, [2,0]} = \\ \mathcal{P}_{D\gamma}^{[2,0]} &= \mathcal{P}_{U\gamma}^{[2,0]} = \mathcal{P}_{L\gamma}^{[2,0]} = \mathcal{P}_{\gamma D}^{[2,0]} = \mathcal{P}_{\gamma U}^{[2,0]} = \mathcal{P}_{\gamma L}^{[2,0]} = \mathcal{P}_{g\gamma}^{[2,0]} = \mathcal{P}_{\gamma g}^{[2,0]} = \mathcal{P}_{\gamma\gamma}^{[2,0]} = 0, \end{aligned} \quad (193)$$

and:

$$\begin{aligned} \mathcal{P}_{DD}^{+, [2,0]} &= \mathcal{P}_{UU}^{+, [2,0]} = P^{+, [1]}, \\ \mathcal{P}_{DD}^{-, [2,0]} &= \mathcal{P}_{UU}^{-, [2,0]} = P^{-, [1]}, \\ \mathcal{P}_{DD}^{\text{PS}, [2,0]} &= \mathcal{P}_{DU}^{\text{PS}, [2,0]} = \mathcal{P}_{DU}^{\text{PS}, [2,0]} = \mathcal{P}_{UU}^{\text{PS}, [2,0]} = P^{\text{PS}, [1]}, \\ \mathcal{P}_{Dg}^{[2,0]} &= \mathcal{P}_{Ug}^{[2,0]} = P_{qg}^{[1]}, \\ \mathcal{P}_{gD}^{[2,0]} &= \mathcal{P}_{gU}^{[2,0]} = P_{gq}^{[1]}, \\ \mathcal{P}_{gg}^{[2,0]} &= P_{gg}^{[1]}. \end{aligned} \quad (194)$$

For the splitting functions at $\mathcal{O}(\alpha^2)$, we have:

$$P^{\text{PV}, [0,2]} = \mathcal{P}_{Dg}^{[0,2]} = \mathcal{P}_{Ug}^{[0,2]} = \mathcal{P}_{gD}^{[0,2]} = \mathcal{P}_{gU}^{[0,2]} = \mathcal{P}_{gg}^{[0,2]} = \mathcal{P}_{g\gamma}^{[0,2]} = \mathcal{P}_{\gamma g}^{[0,2]} = 0, \quad (195)$$

and:

$$\begin{aligned} \mathcal{P}_{XX}^{\pm, [0,2]} &= e_X^4 \left\{ -4 \left[\left(2 \ln(x) \ln(1-x) + \frac{3}{2} \ln(x) \right) \frac{1+x^2}{1-x} + \frac{3+7x}{2} \ln(x) \right. \right. \\ &+ \left. \frac{1+x}{2} \ln^2(x) + 5(1-x) + \left(\frac{\pi^2}{2} - \frac{3}{8} - 6\zeta_3 \right) \delta(1-x) \right] \\ &- \frac{e_\Sigma^2}{e_X^2} 4 \left[\frac{4}{3} (1-x) + \frac{1+x^2}{(1-x)_+} \left(\frac{2}{3} \ln(x) + \frac{10}{9} \right) + \left(\frac{2\pi^2}{9} + \frac{1}{6} \right) \delta(1-x) \right] \\ &\left. \pm 4 \left[4(1-x) + 2(1+x) \ln(x) + 2 \frac{1+x^2}{1+x} S_2(x) \right] \right\}, \\ \mathcal{P}_{XY}^{\text{PS}, [0,2]} &= N_X e_X^2 e_Y^2 8 \left[\frac{20}{9x} - 2 + 6x - \frac{56}{9} x^2 + \left(1 + 5x + \frac{8}{3} x^2 \right) \ln(x) - (1+x) \ln^2(x) \right], \\ \mathcal{P}_{X\gamma}^{[0,2]} &= N_X e_X^4 2 \left[4 - 9x - (1-4x) \ln(x) - (1-2x) \ln^2(x) + 4 \ln(1-x) \right. \\ &+ \left. (x^2 + (1-x)^2) \left(2 \ln^2 \left(\frac{1-x}{x} \right) - 4 \ln \left(\frac{1-x}{x} \right) - \frac{2\pi^2}{3} + 10 \right) \right], \\ \mathcal{P}_{\gamma X}^{[0,2]} &= e_X^4 \left\{ 4 \left[- (3 \ln(1-x) + \ln^2(1-x)) \frac{1+(1-x)^2}{x} + \left(2 + \frac{7}{2} x \right) \ln(x) - \left(1 - \frac{x}{2} \right) \ln^2(x) \right. \right. \\ &- \left. \left. 2x \ln(1-x) - \frac{7}{2} x - \frac{5}{2} \right] - \frac{e_\Sigma^2}{e_X^2} 4 \left[\frac{4}{3} x + \frac{1+(1-x)^2}{x} \left(\frac{20}{9} + \frac{4}{3} \ln(1-x) \right) \right] \right\}, \\ \mathcal{P}_{\gamma\gamma}^{[0,2]} &= e_\Sigma^4 4 \left[-16 + 8x + \frac{20}{3} x^2 + \frac{4}{3x} - (6 + 10x) \ln(x) - 2(1+x) \ln^2(x) - \delta(1-x) \right], \end{aligned} \quad (196)$$

The expressions are taken from Ref. [3].

Let us now move to the $\mathcal{O}(\alpha_s \alpha)$ splitting functions, whose relevant expressions can be read off from Ref. [4]. At this order, we find:

$$\begin{aligned} \mathcal{P}_{LL}^{+,[1,1]} &= \mathcal{P}_{LL}^{-,[1,1]} = \mathcal{P}_{LL}^{\text{PS},[1,1]} = \mathcal{P}_{LU}^{\text{PS},[1,1]} = \mathcal{P}_{LD}^{\text{PS},[1,1]} = \mathcal{P}_{UL}^{\text{PS},[1,1]} = \mathcal{P}_{UU}^{\text{PS},[1,1]} = \\ \mathcal{P}_{UD}^{\text{PS},[1,1]} &= \mathcal{P}_{DL}^{\text{PS},[1,1]} = \mathcal{P}_{DU}^{\text{PS},[1,1]} = \mathcal{P}_{DD}^{\text{PS},[1,1]} = P^{\text{PV}[1,1]} = \mathcal{P}_{L\gamma}^{[1,1]} = \mathcal{P}_{\gamma L}^{[1,1]} = 0, \end{aligned} \quad (197)$$

and:

$$\begin{aligned} \mathcal{P}_{XX}^{\pm,[1,1]} &= -e_X^2 8C_F \left[\left(2\ln(1-x) + \frac{3}{2} \right) \ln(x) \frac{1+x^2}{1-x} + \frac{3+7x}{2} \ln(x) + \frac{1+x}{2} \ln^2(x) \right. \\ &\quad \left. + 5(1-x) + \left(\frac{\pi^2}{2} - \frac{3}{8} - 6\zeta_3 \right) \delta(1-x) \right] \\ &\quad \pm e_X^2 8C_F \left[4(1-x) + 2(1+x) \ln(x) + \frac{1+x^2}{1+x} S_2(x) \right], \quad X = U, D \\ \mathcal{P}_{gX}^{[1,1]} &= e_X^2 4C_F \left[- (3\ln(1-x) + \ln^2(1-x)) \frac{1+(1-x)^2}{x} + \left(2 + \frac{7}{2}x \right) \ln(x) \right. \\ &\quad \left. - \left(1 - \frac{x}{2} \right) \ln^2(x) - 2x \ln(1-x) - \frac{7}{2}x - \frac{5}{2} \right], \quad X = U, D \\ \mathcal{P}_{\gamma X}^{[1,1]} &= \mathcal{P}_{gX}^{[1,1]}, \quad X = U, D \\ \mathcal{P}_{Xg}^{[1,1]} &= e_X^2 2T_R \left[4 - 9x - (1-4x) \ln(x) - (1-2x) \ln^2(x) + 4\ln(1-x) \right. \\ &\quad \left. + (x^2 + (1-x)^2) \left(2\ln^2\left(\frac{1-x}{x}\right) - 4\ln\left(\frac{1-x}{x}\right) - \frac{2\pi^2}{3} + 10 \right) \right], \quad X = U, D \\ \mathcal{P}_{X\gamma}^{[1,1]} &= e_X^2 2N_X C_F \left[4 - 9x - (1-4x) \ln(x) - (1-2x) \ln^2(x) + 4\ln(1-x) \right. \\ &\quad \left. + (x^2 + (1-x)^2) \left(2\ln^2\left(\frac{1-x}{x}\right) - 4\ln\left(\frac{1-x}{x}\right) - \frac{2\pi^2}{3} + 10 \right) \right], \quad X = U, D \\ \mathcal{P}_{gg}^{[1,1]} &= -4T_R (n_D e_D^2 + n_U e_U^2) \delta(1-x), \\ \mathcal{P}_{g\gamma}^{[1,1]} &= 4N_c C_F (n_D e_D^2 + n_U e_U^2) \left[-16 + 8x + \frac{20}{3}x^2 + \frac{4}{3x} - (6+10x) \ln(x) - 2(1+x) \ln^2(x) \right], \\ \mathcal{P}_{\gamma g}^{[1,1]} &= 4T_R (n_D e_D^2 + n_U e_U^2) \left[-16 + 8x + \frac{20}{3}x^2 + \frac{4}{3x} - (6+10x) \ln(x) - 2(1+x) \ln^2(x) \right], \\ \mathcal{P}_{\gamma\gamma}^{[1,1]} &= -4N_c C_F (n_D e_D^2 + n_U e_U^2) \delta(1-x). \end{aligned} \quad (198)$$

At higher orders, where only QCD corrections are included, one has that all splitting functions involving leptons and/or photon are zero:

$$\begin{aligned} \mathcal{P}_{LL}^{+,[n,0]} &= \mathcal{P}_{LL}^{-,[n,0]} = \mathcal{P}_{LL}^{\text{PS},[n,0]} = \mathcal{P}_{LU}^{\text{PS},[n,0]} = \mathcal{P}_{LD}^{\text{PS},[n,0]} = \mathcal{P}_{UL}^{\text{PS},[n,0]} = \mathcal{P}_{UL}^{\text{PS},[n,0]} = \mathcal{P}_{DL}^{\text{PS},[n,0]} = \\ \mathcal{P}_{D\gamma}^{[n,0]} &= \mathcal{P}_{U\gamma}^{[n,0]} = \mathcal{P}_{L\gamma}^{[n,0]} = \mathcal{P}_{\gamma D}^{[n,0]} = \mathcal{P}_{\gamma U}^{[n,0]} = \mathcal{P}_{\gamma L}^{[n,0]} = \mathcal{P}_{g\gamma}^{[n,0]} = \mathcal{P}_{\gamma g}^{[n,0]} = \mathcal{P}_{\gamma\gamma}^{[n,0]} = 0, \end{aligned} \quad (199)$$

for $n \geq 3$. The others are given by the QCD ones as follows:

$$\begin{aligned}
\mathcal{P}_{DD}^{\pm,[n,0]} &= \mathcal{P}_{UU}^{\pm,[n,0]} = P^{\pm,[n-1]}, \\
\mathcal{P}_{UU}^{\text{PS},[n,0]} &= \mathcal{P}_{UD}^{\text{PS},[n,0]} = \mathcal{P}_{DU}^{\text{PS},[n,0]} = \mathcal{P}_{DD}^{\text{PS},[n,0]} = P^{\text{PS},[n-1]}, \\
\mathcal{P}_{UV}^{\text{PV},[n,0]} &= P^{\text{PV},[n-1]}, \\
\mathcal{P}_{Dg}^{[n,0]} &= \mathcal{P}_{Ug}^{[n,0]} = P_{gg}^{[n-1]}, \\
\mathcal{P}_{qD}^{[n,0]} &= \mathcal{P}_{qU}^{[n,0]} = P_{gq}^{[n-1]}, \\
\mathcal{P}_{gg}^{[n,0]} &= P_{gg}^{[n-1]},
\end{aligned} \tag{200}$$

where $P^{[n-1]}$ denote the QCD splitting functions at $N^{n-1}\text{LO}$.

We can finally move to the matching conditions. Looking at Eq. (185), starting from NLO accuracy, we need specify the various functions $\mathcal{K}$. At each heavy-quark threshold, the functions $\mathcal{K}$ also depend on the ration $\kappa_X = \mu_X/m_X$, where μ_X and m_X are the threshold and the mass of the quarks that is becoming active. We have a total of 12 functions to determine:

$$\mathcal{K}_{Qg}, \mathcal{K}_{X\gamma}, \mathcal{K}_{Qq'}, \mathcal{K}_{XX}, \mathcal{K}_{qg}, \mathcal{K}_{\text{NS},q}, \mathcal{K}_{qq'}, \mathcal{K}_{gg}, \mathcal{K}_{gq}, \mathcal{K}_{gQ}, \mathcal{K}_{\gamma\gamma}^X, \mathcal{K}_{\gamma X}, \tag{201}$$

As already discussed above, if X is a lepton ($X = L$), many of these vanish (see Eq. (186)). If instead X is either a down- or a up-type quark ($X = D, U$), the functions coincide with the pure-QCD ones. Specifically, referring to Eq. (154), one has:

$$\begin{aligned}
\mathcal{K}_{Qg} &= \begin{cases} K_{hg} & X = D, U, \\ 0 & X = L, \end{cases} \\
\mathcal{K}_{Qq'} &= \begin{cases} K_{hl} & X = D, U, \\ 0 & X = L, \end{cases} \\
\mathcal{K}_{qg} &= \begin{cases} K_{lg} & X = D, U, \\ 0 & X = L, \end{cases} \\
\mathcal{K}_{\text{NS},q} &= \begin{cases} K_{ll} & X = D, U, \\ 0 & X = L, \end{cases} \\
\mathcal{K}_{qq'} &= \begin{cases} K_{ll'} & X = D, U, \\ 0 & X = L, \end{cases} \\
\mathcal{K}_{gg} &= \begin{cases} K_{gg} & X = D, U, \\ 0 & X = L, \end{cases} \\
\mathcal{K}_{gq} &= \begin{cases} K_{gl} & X = D, U, \\ 0 & X = L, \end{cases} \\
\mathcal{K}_{gQ} &= \begin{cases} K_{gh} & X = D, U, \\ 0 & X = L, \end{cases} \\
\mathcal{K}_{\text{NS},q}^- &= \begin{cases} K_{ll}^- & X = D, U, \\ 0 & X = L. \end{cases}
\end{aligned} \tag{202}$$

The remaining are:

$$\begin{aligned}
\mathcal{K}_{XX} &= \begin{cases} K_{hh} & X = D, U, \\ 2e_X^2 \left[\frac{1+x^2}{1-x} (\ln(\kappa_X^2) - 1 - 2\ln(1-x)) \right]_+ & X = L, \end{cases} \\
\mathcal{K}_{X\gamma} &= 4N_X e_X^2 [x^2 + (1-x)^2] \ln(\kappa_X^2), \\
\mathcal{K}_{\gamma X} &= 2e_X^2 \left[\frac{1+(1-x)^2}{x} \right] [\ln(\kappa_X^2) - 1 - 2\ln(x)], \\
\mathcal{K}_{\gamma\gamma}^X &= -\frac{4}{3} e_X^2 \ln(\kappa_X^2) \delta(1-x).
\end{aligned} \tag{203}$$

4.1 Sum rules

One of the possible check the implementation of the correctness of splitting functions is that sum rules have to be fulfilled at all scales order by order in perturbation theory. This check is particularly useful when including QED corrections given the proliferation of splitting functions. Sum rules can be checked numerically either at the level of evolved PDFs (as long as they fulfil them at the initial scale) or at the level of splitting functions.

Let us start with the momentum sum rule, which at the level of PDFs, prescribes that:

$$\int_0^1 dy y \left[\sum_i D_i^+(y, \mu) + \sum_j U_j^+(y, \mu) + \sum_\alpha L_\alpha^+(y, \mu) + g(y, \mu) + \gamma(y, \mu) \right] = 1, \tag{204}$$

for all values of μ . Taking the derivative w.r.t. μ of this equality and using the evolution equations allows us to derive integral equations which must be obeyed by splitting functions order by order in perturbation theory. Specifically, we find:

$$\begin{aligned}
\int_0^1 dy y \left[\mathcal{P}_{LL}^{+, [a, b]}(y) + n_L \mathcal{P}_{LL}^{\text{PS}, [a, b]}(y) + n_U \mathcal{P}_{UL}^{\text{PS}, [a, b]}(y) + n_D \mathcal{P}_{DL}^{\text{PS}, [a, b]}(y) + \mathcal{P}_{\gamma L}^{[a, b]}(y) \right] &= 0, \\
\int_0^1 dy y \left[\mathcal{P}_{UU}^{+, [a, b]}(y) + n_L \mathcal{P}_{LU}^{\text{PS}, [a, b]}(y) + n_U \mathcal{P}_{UU}^{\text{PS}, [a, b]}(y) + n_D \mathcal{P}_{DU}^{\text{PS}, [a, b]}(y) + \mathcal{P}_{gU}^{[a, b]}(y) + \mathcal{P}_{\gamma U}^{[a, b]}(y) \right] &= 0, \\
\int_0^1 dy y \left[\mathcal{P}_{DD}^{+, [a, b]}(y) + n_L \mathcal{P}_{LD}^{\text{PS}, [a, b]}(y) + n_U \mathcal{P}_{UD}^{\text{PS}, [a, b]}(y) + n_D \mathcal{P}_{DD}^{\text{PS}, [a, b]}(y) + \mathcal{P}_{gD}^{[a, b]}(y) + \mathcal{P}_{\gamma D}^{[a, b]}(y) \right] &= 0, \\
\int_0^1 dy y \left[2n_U \mathcal{P}_{Ug}^{[a, b]}(y) + 2n_D \mathcal{P}_{Dg}^{[a, b]}(y) + \mathcal{P}_{\gamma g}^{[a, b]}(y) + \mathcal{P}_{gg}^{[a, b]}(y) \right] &= 0, \\
\int_0^1 dy y \left[2n_L \mathcal{P}_{L\gamma}^{[a, b]}(y) + 2n_U \mathcal{P}_{U\gamma}^{[a, b]}(y) + 2n_D \mathcal{P}_{D\gamma}^{[a, b]}(y) + \mathcal{P}_{g\gamma}^{[a, b]}(y) + \mathcal{P}_{\gamma\gamma}^{[a, b]}(y) \right] &= 0,
\end{aligned} \tag{205}$$

for the any of the perturbative orders $[a, b]$ of our interest. For the leading-order sets of splitting functions, it is straightforward to check analytically that this equalities are true. Beyond these orders analytic checks are still possible but more cumbersome.

We next consider the valence (or counting) sum rule, which at the level of PDFs reads:

$$\int_0^1 dy X_i^-(y, \mu) = \text{constant}, \quad X = D, U, L, \tag{206}$$

where the constant depends on X . At the level of splitting functions, the valence sum rule implies:

$$\begin{aligned} \int_0^1 dy \left[\mathcal{P}_{XX}^{-,[a,b]}(y) + n_X P^{\text{PV},[a,b]}(y) \right] &= 0, \quad X = D, U, \\ \int_0^1 dy \mathcal{P}_{LL}^{-,[a,b]}(y) &= 0. \end{aligned} \tag{207}$$

Also matching functions are subject to integral identities deriving from the sum rules. Indeed, matching functions must be such that valence and momentum sum rules are preserved when a threshold is crossed. In pure QCD, we found the identities in Eq. (157). In the presence of QED corrections, we refer to the system in Eq. (185) and from the valence sum rule we derive:

$$\int_0^1 dy \mathcal{K}_{\text{NS},q}^{-,[a,b]}(y) = 0, \tag{208}$$

while from the momentum sum rule we find:

$$\begin{aligned} \int_0^1 dy y \left[\mathcal{K}_{Qg}^{[a,b]}(y) + n_f \mathcal{K}_{qg}^{[a,b]}(y) + \mathcal{K}_{gg}^{[a,b]}(y) \right] &= 0, \\ \int_0^1 dy y \left[\mathcal{K}_{X\gamma}^{[a,b]}(y) + \mathcal{K}_{\gamma\gamma}^{X,[a,b]}(y) \right] &= 0, \\ \int_0^1 dy y \left[\mathcal{K}_{Qq'}^{[a,b]}(y) + \mathcal{K}_{\text{NS},q}^{[a,b]}(y) + n_f \mathcal{K}_{qq'}^{[a,b]}(y) + \mathcal{K}_{gq}^{[a,b]}(y) \right] &= 0, \\ \int_0^1 dy y \left[\mathcal{K}_{XX}^{[a,b]}(y) + \mathcal{K}_{gQ}^{[a,b]}(y) + \mathcal{K}_{\gamma X}^{[a,b]}(y) \right] &= 0, \end{aligned} \tag{209}$$

where we defined $n_f = n_D + n_U$.

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